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# Long-term durability of basalt- and glass-fibre reinforced polymer (BFRP/GFRP) bars in seawater and sea sand concrete environment

Zike Wang [a,b], Xiao-Ling Zhao [b,*], Guijun Xian [a], Gang Wu [c], R.K. Singh Raman [d,e], Saad Al-Saadi [d], Asadul Haque [b]

[a] School of Civil Engineering, Harbin Institute of Technology, Harbin 150090, China
[b] Department of Civil Engineering, Monash University, Clayton, VIC 3800, Australia
[c] International Institute for Urban Systems Engineering, Southeast University, Nanjing 210096, China
[d] Department of Mechanical and Aerospace Engineering, Monash University, Clayton, VIC 3800, Australia
[e] Department of Chemical Engineering, Monash University, Clayton, VIC 3800, Australia

## HIGHLIGHTS

- Strength tests on BFRP/GFRP bars in seawater and sea sand concrete (SWSSC) environment.
- GFRP bars are more durable than BFRP bars in SWSSC environment.
- No change in Young's modulus for GFRP and BFRP bars after exposure in SWSSC solutions.
- Current models are conservative in predicting long-term strength of BFRP/GFRP bars.

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## ABSTRACT

This paper presents a study on the long-term performance of basalt- and glass-fibre reinforced polymer (BFRP/GFRP) bars in seawater and sea sand concrete (SWSSC) environment. Accelerated corrosion tests were conducted using two types of SWSSC solutions at different pH and temperatures, and for different durations. The tensile tests of pre-exposed bars suggested the GFRP bars to be more durable than BFRP bars, while the Young's modulus of all specimens remained unchanged. Scanning electron microscopy (SEM), X-ray computed tomography (CT) and energy dispersive X-ray spectroscopy (EDS) results were utilized to explain the damage mechanism. The long-term behaviour of BFRP and GFRP bars under the service construction condition was also predicted using Arrhenius degradation theory.



## 1. Introduction

In responding to the resource shortage caused by the global population increase, the seawater and sea sand concrete (SWSSC) was recently proposed to replace ordinary Portland cement (OPC)-based concrete [1–3]. Compared with normal concrete, SWSSC utilizes seawater, sea sand to replace freshwater, river sand in OPC. In addition, SWSSC also can use industry wastes (e.g. slag, fly ash). Another benefit of SWSSC is the reduced probability of concrete cracking that can be caused by alkali silica reaction (ASR) [4–6]. In addition, the mechanical properties of SWSSC are generally similar to those of conventional Portland concrete [6].

Fibre reinforced polymer (FRP) composites are increasingly used in civil engineering because of their excellent corrosion resistance and high strength-to-weight ratio. When used together with concrete, FRP composites can act as either external confinement (e.g. concrete-filled FRP tubes) or internal reinforcement. Studies on SWSSC-filled FRP tubes (either fully filled or double skin arrangement) have been reported [1,7]. In the present work, the durability of FRP bars as internal reinforcement in SWSSC environment is investigated.

FRP bars are used to replace steel bars in reinforced concrete (RC) structures to overcome the corrosion problem of steel bars. The traditional FRP bars generally refer to glass- and carbon- FRP bars, with GFRP bars more widely used due to their low cost [8]. Recently, basalt fibre reinforced polymer (BFRP) has been developed, which is even cheaper [9]. Basalt fibres are an

* Corresponding author.
*E-mail address:* ZXL@monash.edu (X.-L. Zhao).

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environmentally friendly and non-hazardous material as they are directly produced from volcanic rocks without additives [10].

The durability of FRP bars (including GFRP, CFRP and BFRP) in civil engineering environment, especially in normal concrete environment has been studied by many researchers [8,11–20]. Previous studies revealed that the important factors influencing the degradation rate of GFRP bars include pH [18], exposure temperature [20], moisture content of concrete [12,14,16], polymer type [11] and pre-loading level [8,15,21]. The degradation mechanism of GFRP bars is discussed in the published literatures [22–25]. However, only limited work [13,17,26,27] was reported on the durability of BFRP bars. Wu et al. [13,27] found that the tensile strength retention of BFRP bars in alkaline concrete pore solution reduced much quicker than that in deionized water, salt and acid environments. Serbescu et al. suggested that exposure temperature was the most important factor affecting the degradation process of BFRP bars [26]. More recently, Benmokrane et al. studied the effect of polymer type on durability of BFRP bars and the results showed that the basalt/epoxy FRP bars had better durability than basalt/vinyl ester FRP bars in alkaline solution [17].

However, to the best of authors' knowledge, only limited experimental studies were recently carried out on the bond durability between steel-FRP composite bars and sea sand concrete [19,28]. There is a lack of study on the durability of BFRP and GFRP bars

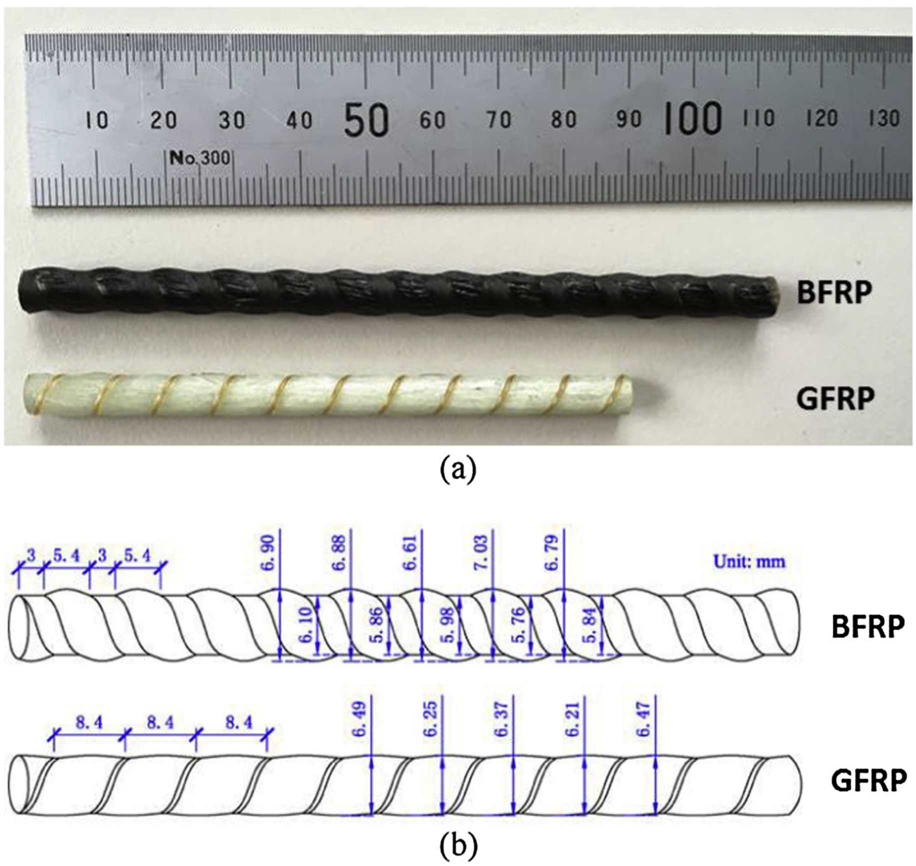



**Fig. 1.** Tested FRP bars: (a) photographs and (b) schematic diagrams.

**Table 1**
Production parameters of FRP bars.

| Type | Nominal diameter (mm) | Reinforced fibre | | | | | Resin matrix | | $V_f$ (%) |
|---|---|---|---|---|---|---|---|---|---|
| | | Brand | Diameter (μm) | Tensile strength (MPa) | Tensile modulus (GPa) | Density (g/cm$^3$) | Epoxy | Tg (°C) | |
| BFRP | 6 | 1200 Tex | 13 | 3800–4840 | 93.1–110 | 2.65 | Bisphenol-A E-51 | ~130 | ~65 |
| GFRP | 6 | E-glass 9600 Tex | 28 | 2200 | 76 | 2.54 | Bisphenol-A E-51 | 140 | ~62.8 |

**Table 2**
Compositions of simulated SWSSC pore solutions.

| Type | Quantities (gram per litre) | | | | pH |
|---|---|---|---|---|---|
| | NaOH | KOH | $Ca(OH)_2$ | NaCl | |
| N-SWSSC solution | 2.4 | 19.6 | 2 | 35 | 13.4 |
| HP-SWSSC solution | 0.6 | 1.4 | 0.037 | 35 | 12.7 |

under the SWSSC environment. This paper aims to fill this knowledge gap by investigating the tensile properties of BFRP and GFRP bars within SWSSC environment using the accelerated corrosion tests.

In this study, two types of simulated SWSSC pore solutions are used, namely normal seawater and sea sand concrete (N-SWSSC) solution and high performance seawater and sea sand concrete (HP-SWSSC) solution. Four exposure temperatures (room temperature, 40, 48 and 55 °C) were adopted in the testing program. After exposure to N-SWSSC and HP-SWSSC solutions for 21, 42 and 63 days, respectively, the tensile strength and Young's modulus of each specimen were determined. In addition, to investigate the damage mechanism of GFRP and BFRP bars, scanning electron microscopy (SEM), X-ray computed tomography (CT) and energy dispersive X-ray spectroscopy (EDS) methods were used to characterize the changes of microstructure and chemical elements for BFRP and GFRP bars after exposure. Finally, the long-term behaviour of BFRP and GFRP bars under simulated civil environment was predicted with the Arrhenius relation theory. The predicted results are compared with the reported field observation data [29].

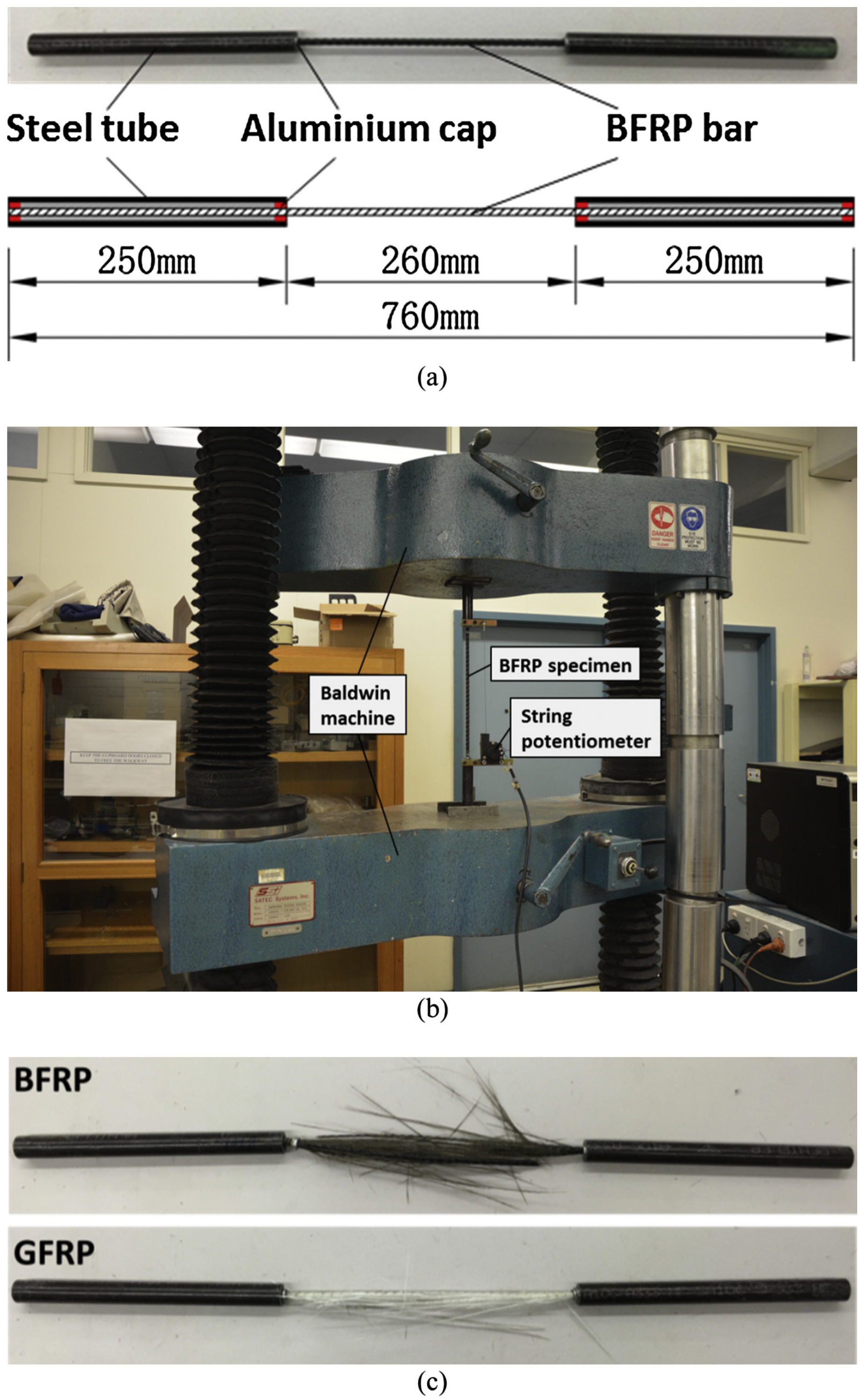



**Fig. 2.** Tensile specimens: (a) reference/intact specimen; (b) typical FRP bars under test and (c) specimens after test.

## 2. Experimental program

### 2.1. Materials

The unidirectional BFRP and GFRP bars with the normal diameter of 6 mm are used in this study and presented in Fig. 1, and the BFRP bars were identical to those in [13]. This paper studies the durability of BFRP bars within seawater sea sand concrete environment. The reason to select BFRP bars with a diameter of 6 mm is to have a direct comparison with the test results given in Ref. [13] where normal concrete environment was adopted. BFRP and GFRP bars were both produced using the same epoxy resin and corresponding reinforced fibres by pultrusion. The basalt fibres were manufactured by Zhejiang GBF Basalt Fibre Co. Ltd. (Hengdian, China). The E-glass fibres were manufactured by Chongqing Polymer International Corp. (Chongqing, China). The epoxy resin was procured from Bluestar New Chemical Materials Co. Ltd (Wuxi, China). Based on the information from suppliers, the molding temperatures in six zones for BFRP during pultrusion were from 160 to 300 °C, the pull through speed was 150 cm/min, and the basalt fibre volume fraction ($V_f$) is approximately 65%. While the molding temperature was kept as 140 °C in all three zones for GFRP during pultrusion, the pull through speed was 40 cm/min, and the glass fibre volume fraction ($V_f$) is approximately 62.8%. The different curing parameters caused the difference in glass transition temperatures for the cured epoxy resin inside BFRP and GFRP. All the production parameters of BFRP and GFRP bars provided by manufactures are shown in Table 1.

As shown in Fig. 1(b), the surface of BFRP bars was ribbed with a nylon laminate during pultrusion. The surface of GFRP bars was helically wrapped with a Kevlar fibre bundle with 0.4 mm diameter. These rough treatments cause that the diameter is not uniform along the surface of FRP bars. The measured average outer diameter of BFRP is 6.82 ± 0.17 mm and the inner diameter is 5.89 ± 0.15 mm. For GFRP bars, the outer and inner diameters are rather close, and the measured average outer diameter is 6.33 ± 0.10 mm. For the calculation of cross-section area of FRP bars, the average diameter was adopted, i.e. 6.0 mm for BFRP and 6.3 mm for GFRP.

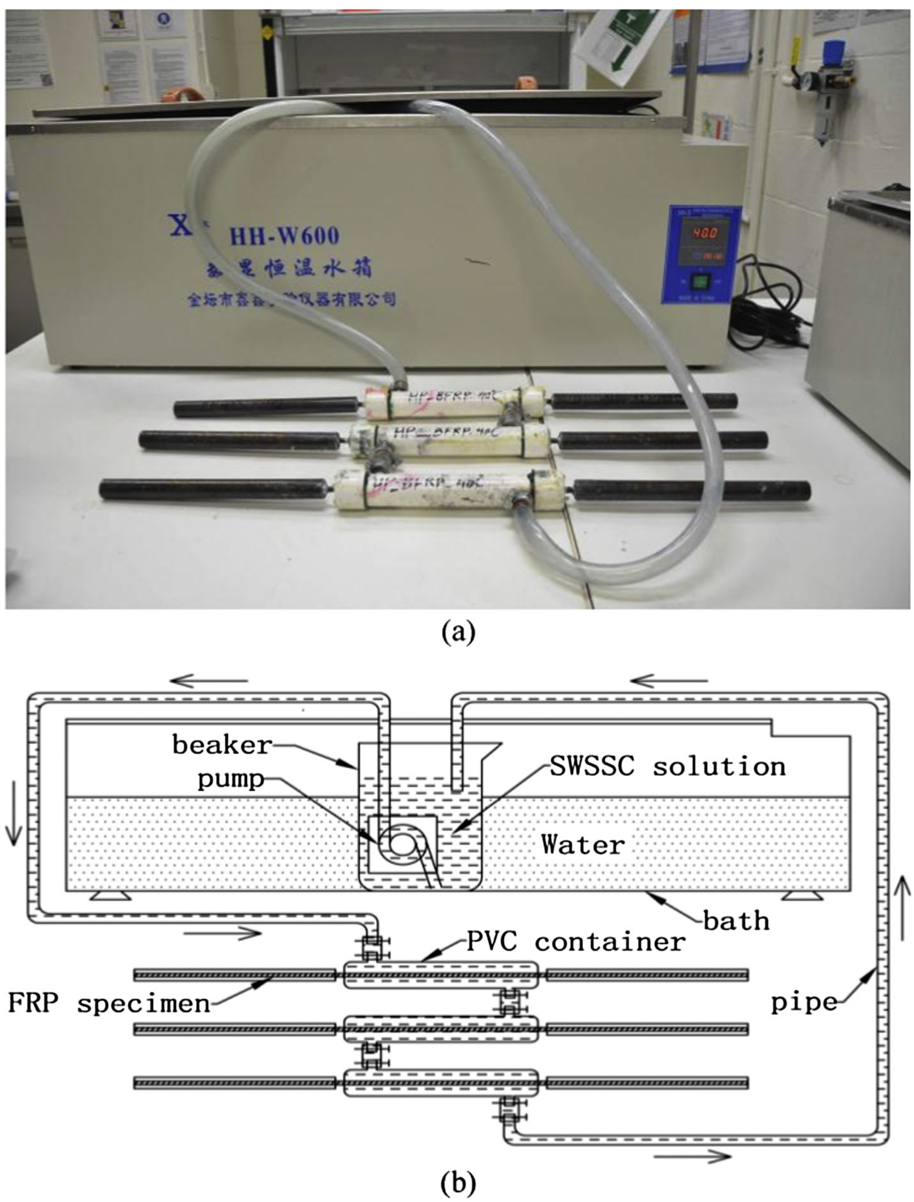



**Fig. 3.** SWSSC solution exposure setup: (a) photograph and (b) schematic diagram.

### 2.2. Simulated seawater and sea sand concrete (SWSSC) solution

As shown in Table 2, two kinds of simulated SWSSC pore solutions were chosen in this study. The normal seawater sea sand concrete (N-SWSSC) solution was prepared by mixing: sodium hydroxide (2.4 g/l, NaOH), potassium hydroxide (19.6 g/l, KOH), calcium hydroxide (2 g/l, $Ca(OH)_2$), sodium chloride (35 g/l, NaCl) and distilled water (to make the volume of solution of one litre). Similarly, the high performance seawater sea sand concrete (HP-SWSSC) solution was prepared by mixing: sodium hydroxide (0.6 g/l, NaOH), potassium hydroxide (1.4 g/l, KOH), calcium hydroxide (0.037 g/l, $Ca(OH)_2$), sodium chloride (35 g/l, NaCl) and distilled water (to make the volume of solution of one litre). The mass concentrations of NaOH, KOH and $Ca(OH)_2$ were chosen according to Chen et al. [12] to simulate the normal concrete and high performance concrete pore solutions. The pH of N-SWSSC and HP-SWSSC solutions were measured to be 13.4 and 12.7, respectively. The molar concentrations of chemicals (NaOH, KOH, $Ca(OH)_2$ and NaCl) are used to prepare the simulated SWSSC pore solutions which produce alkali ions/elements with concentrations close to those produced in real seawater and sea sand concrete pore solution.

### 2.3. Tensile test

The tensile tests of reference and conditioned BFRP and GFRP bars were conducted in accordance with ACI-440.3R B.2. The length of the tensile specimen was 760 mm. Each end was coated with silica sands and then mounted in 250 mm-long steel tubes grouted with an epoxy resin mortar matrix. Aluminium caps were used to close both ends of the steel tubes and to keep the bar in the centre of the tube during anchor installation. The reference BFRP specimen is shown in Fig. 2(a). The length of the test section was 260 mm. As shown in Fig. 2(b), the test was carried out by gripping the steel tube into the wedges of a Baldwin machine that has a capacity of 500 kN. A loading rate of 2 mm/min was adopted and the string potentiometer was used to measure the deformation of FRP bars. For BFRP and GFRP bars, three reference specimens were tested (with two conditioned specimens tested for each group). The typical reference BFRP and GFRP failed specimens after test are shown in Fig. 2(c).

As shown in Fig. 3, a solution circulating system was designed for the exposure experiment. The middle part of the tensile specimen was inserted inside a 30 mm-diameter and 240 mm-long PVC pipe which was used as the chamber for exposure to the simulated SWSSC solution. Both ends of the PVC pipe (container) were sealed with epoxy adhesive that ensured no leakage during conditioning. The SWSSC solution was contained in a beaker with 5 L capacity that was placed in the thermostatic water bath to maintain the temperature. A submerged pump was used to circulate/transport the SWSSC solution through the whole circulating system. Four exposure temperatures (room temperature, 40, 48 and 55 °C) were adopted for BFRP specimens and three exposure temperatures (room temperature, 40 and 55 °C) for GFRP specimens. After exposure in SWSSC solution for 21, 42 and 63 days at each temperature, BFRP and GFRP specimens were removed from the aging system and tested under tension to compare their tensile strength with that of the reference specimen. All the pre-exposure experiments were conducted in the indoor environment. The room temperature of two kinds of SWSSC solutions at room temperature condition during the whole exposure period was monitored to be 32.2 ± 1.2 °C. Accordingly, the "room temperature" in this study refers to 32 °C.

In this study, a total of 45 BFRP and 39 GFRP tensile specimens were tested, and the specimen label system was also defined as follows:

For unconditioned specimens, "BR" represents the BFRP reference specimen and "GR" represents the GFRP reference specimen.

For conditioned specimens, "NBT32D21" represents the BFRP specimen after pre-exposure in N-SWSSC solution at 32 °C for 21 days. In the label "NBT32D21", "N" means the N-SWSSC solution; "B" means the BFRP bar; "T32" means that the exposure temperature is 32 °C and "D21" means that the durability time is 21 days. Similarly, "HPGT32D21" represents the GFRP specimen after pre-exposure in HP-SWSSC solution at 32 °C for 21 days.

For the data referred to from the published literature [13], "ABT25" means the BFRP specimens after pre-exposure in alkaline solution [13] at 25 °C. In the label "ABT25", "A" means the alkaline solution used in the literature [13], "B" means the BFRP bar and "T25" means the exposure temperature is 25 °C.

**Table 3**
Experimental tensile properties of reference and conditioned BFRP specimens.

| Specimen label | Tensile strength | | | Elastic modulus | |
|---|---|---|---|---|---|
| | Mean (MPa) | Retention (%) | CV[a] (%) | Mean (GPa) | CV (%) |
| BR | 1351 | 100 | 5.5 | 39 | 4.7 |
| NBT32D21 | 1317 | 97.5 | 2.1 | 44 | 3.3 |
| NBT32D42 | 1250 | 92.5 | 6.0 | 40 | – |
| NBT32D63 | 1253 | 92.7 | 8.0 | – | – |
| NBT40D21 | 1273 | 94.2 | – | 43 | – |
| NBT40D42 | 1199 | 88.7 | 1.2 | – | – |
| NBT40D63 | 1103 | 81.7 | 0.9 | – | – |
| NBT48D21 | 1257 | 93.0 | 0.6 | 42 | 5.2 |
| NBT48D42 | 1113 | 82.4 | 4.1 | 42 | 11.1 |
| NBT48D63 | 799 | 59.1 | – | 46 | 3.6 |
| NBT55D21 | 908 | 67.2 | 9.0 | 40 | – |
| NBT55D42 | 507 | 37.5 | 4.6 | 36 | – |
| NBT55D63 | 352 | 26.0 | 0.3 | 39 | 1.5 |
| HPBT32D21 | 1341 | 99.3 | 6.5 | 46 | 1.0 |
| HPBT32D42 | 1267 | 93.8 | 0.7 | 39 | 15.5 |
| HPBT32D63 | 1323 | 97.9 | 4.3 | 45 | – |
| HPBT40D21 | 1288 | 95.3 | 2.9 | 43 | 2.2 |
| HPBT40D42 | 1256 | 93.0 | 1.1 | 39 | 12.0 |
| HPBT40D63 | 1219 | 90.2 | 0.1 | 46 | 3.3 |
| HPBT55D21 | 1212 | 89.7 | 3.8 | 44 | 2.2 |
| HPBT55D42 | 1096 | 81.1 | 0.8 | 43 | 0.2 |
| HPBT55D63 | 1046 | 77.4 | 3.7 | 47 | 1.0 |

Note: a: Coefficient of variation.
"–" represents "not available".

**Table 4**
Experimental tensile properties of reference and conditioned GFRP specimens.

| Specimen label | Tensile strength | | | Elastic modulus | |
|---|---|---|---|---|---|
| | Mean (MPa) | Retention (%) | CV[a] (%) | Mean (GPa) | CV (%) |
| GR | 1059 | 100 | 2.5 | 25 | 15.8 |
| NGT32D21 | 952 | 89.9 | 5.7 | – | – |
| NGT32D42 | 966 | 91.2 | 0.7 | – | – |
| NGT32D63 | 925 | 87.4 | 5.4 | – | – |
| NGT40D21 | 903 | 85.3 | 2.6 | 34 | – |
| NGT40D42 | 951 | 89.9 | 0.4 | – | – |
| NGT40D63 | 961 | 90.8 | 0.8 | – | – |
| NGT55D21 | 855 | 80.8 | 13.6 | 31 | – |
| NGT55D42 | 728 | 68.7 | 13.4 | – | – |
| NGT55D63 | 848 | 80.1 | 2.4 | – | – |
| HPGT32D21 | 1032 | 97.5 | 2.7 | 33 | 1.9 |
| HPGT32D42 | 1007 | 95.2 | 1.1 | 34 | 2.5 |
| HPGT32D63 | 1036 | 97.9 | 1.5 | 34 | 0.2 |
| HPGT40D21 | 959 | 90.6 | 10.5 | 33 | 2.4 |
| HPGT40D42 | 962 | 90.9 | 0.4 | 34 | 2.1 |
| HPGT40D63 | 996 | 94.1 | 1.1 | 34 | 1.4 |
| HPGT55D21 | 966 | 91.3 | 3.6 | 33 | 0.3 |
| HPGT55D42 | 980 | 92.6 | 3.0 | 33 | 2.6 |
| HPGT55D63 | 948 | 89.6 | 7.3 | 33 | 1.5 |

Note: a: Coefficient of variation.
"–" represents "not available".

### 2.4. Scanning electron microscopy (SEM) and energy dispersive X-ray spectroscopy (EDS)

The morphologies of the cross sections of the reference and conditioned BFRP and GFRP specimens were characterized with a JEOL 7001F field emission scanning electron microscope (SEM, Japan). In addition, energy dispersive X-ray spectroscopy (EDS) was used to analyse the broad chemical compositions on the cross-section surface of GFRP and BFRP specimen, with the Oxford Instruments AZtec software.

### 2.5. X-ray computed tomography (CT)

In recent years, X-ray computed tomography (CT) scanning technology has become an important diagnostic tool to characterise microstructure and failure mechanism in particulate and long fibre reinforced composites [30]. Compared with the conventional optical, scanning electron microscopy (SEM) and ultrasonic inspection methods, X-ray CT possesses special advantages, such as non-destructive features, 3D imaging and excellent resolution. Therefore, X-ray CT is frequently utilized to observe the fibre in composites [31–33]. To the authors' best knowledge, X-ray CT has not been used to study the distribution of defects (delamination, cracking and micro-voids) in GFRP or BFRP bars used in civil environment.

In this study, the 3D structures of reference and conditioned BFRP and GFRP specimens were imaged using an ultra-high resolution (0.7 μm, Zeiss Xradia XRM520Versa) X-ray Microscopy Facility for Imaging Geo-materials (XMFIG) [34] in the Department of Civil Engineering of Monash University, Australia. The XMFIG has been established through an Australian Research Council Linkage Infrastructure and Equipment Fund (LE130100006) [35]. Scans were performed on the small pieces of FRP specimens (~1 mm diameter) chosen from the edge part of FRP bars. The scans were carried out using a source voltage of 80 kV and power of 7 W at voxel sizes of 0.97–2.25 μm for BFRP and 0.97–1.08 μm for GFRP. In the scanning process, 801 projections over 180 degrees rotation were acquired. The projection images were reconstructed using the XMReconstructor software [36]. The reconstructed CT images were analysed with the Drishti (v2.6.2) software. The image segmentation was performed manually using the histogram. Through image analysis, the distribution and volume contents of various constitutes (i.e., fibre, resin and void) of the FRP specimens were obtained.

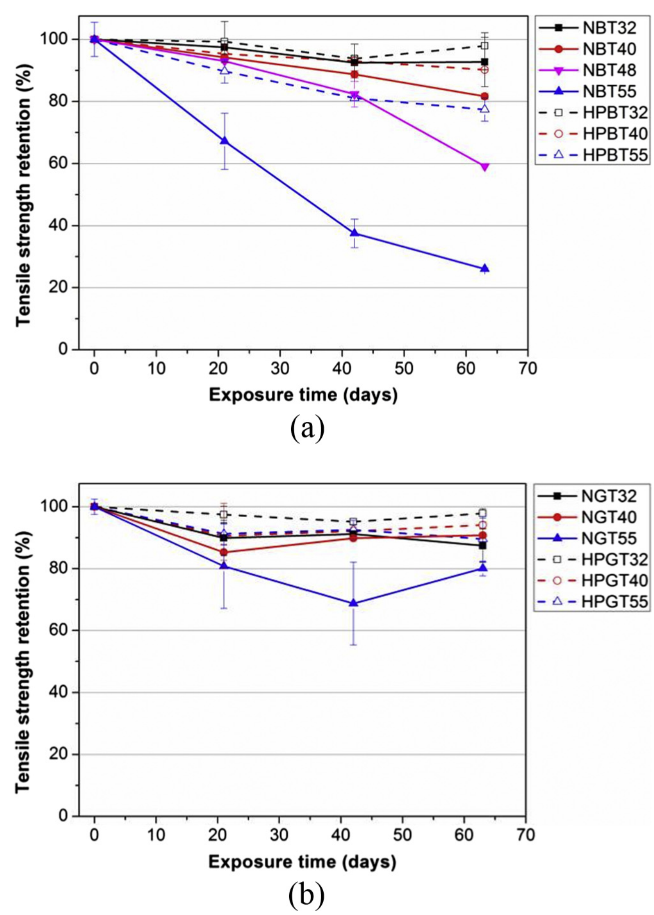



**Fig. 4.** Effect of conditioning in SWSSC solutions on the tensile strength retention of FRP bars: (a) BFRP and (b) GFRP.

## 3. Results and discussion

### 3.1. Tensile properties results

All the tensile test results of BFRP and GFRP specimens are summarized in Tables 3 and 4, respectively. The tensile strength of reference BFRP specimens was determined to be 1351 ± 74 MPa, which is close to the reported data (1398 ± 30 MPa) [13]. The tensile strength of reference GFRP specimens was determined to be 1059 ± 26 MPa, (i.e., significantly lower than that of BFRP bars). Moreover, the tensile elastic modulus of reference GFRP is lower than that of reference BFRP, which can be attributed to the lower strength and modulus of glass fibre than those of basalt fibre.

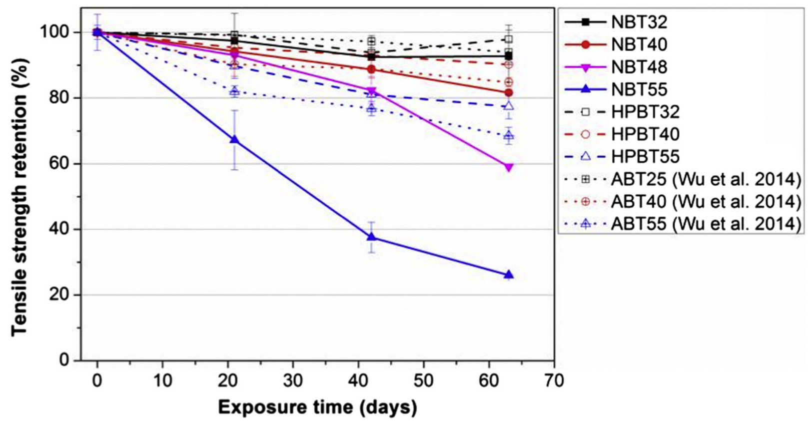



**Fig. 5.** Comparison of tensile strength retention of BFRP bars in SWSSC and alkaline environments at different temperatures.

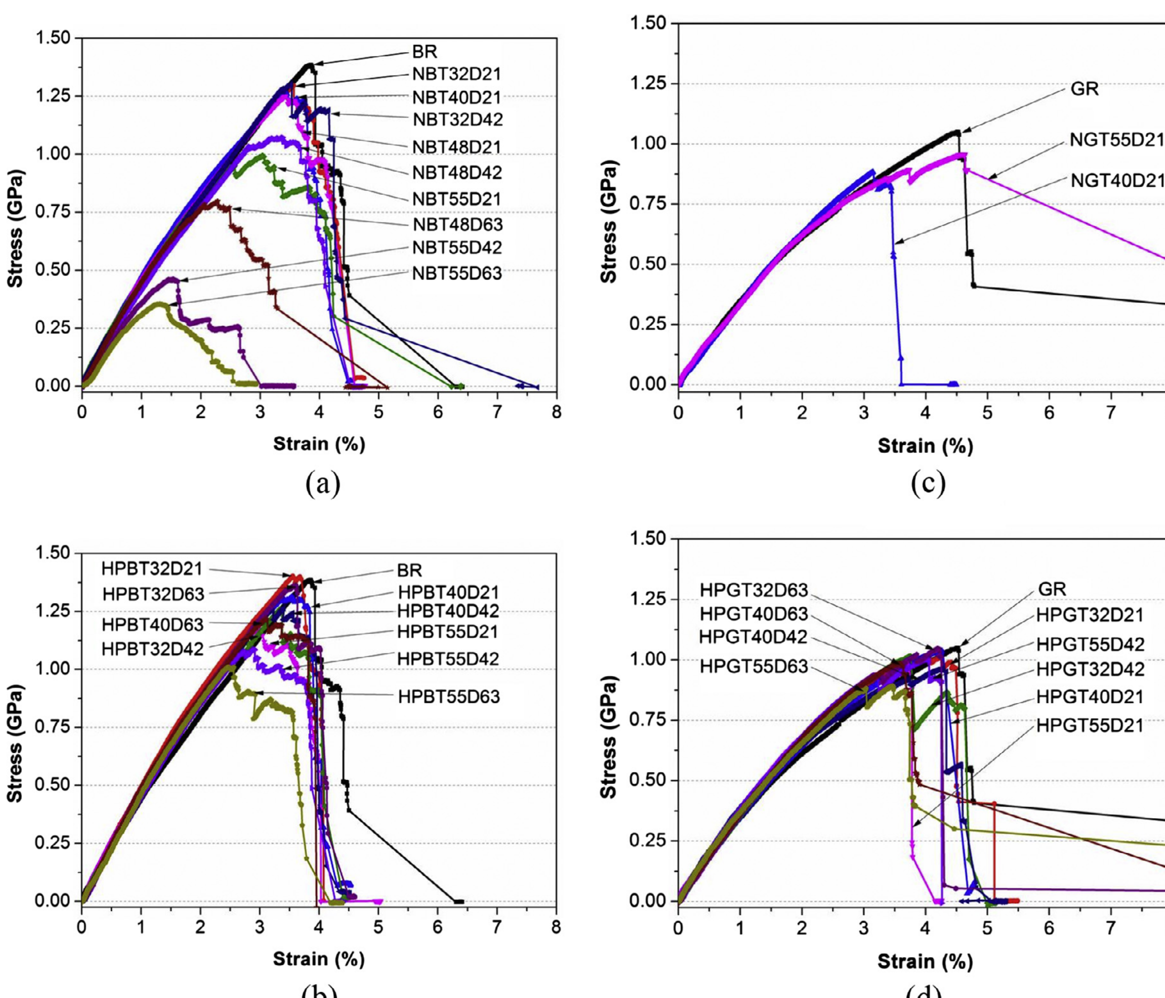



**Fig. 6.** Effect of conditioning in SWSSC solutions on tensile stress-strain curves of FRP bars: (a) BFRP in N-SWSSC solution; (b) BFRP in HP-SWSSC solution; (c) GFRP in N-SWSSC solution and (d) GFRP in HP-SWSSC solution.

#### 3.1.1. Tensile strength retention

The two SWSSC solutions caused damages to the BFRP and GFRP bars. The tensile retention results of BFRP and GFRP specimens pre-exposed to the two solutions at different temperatures are compared in Fig. 4(a) and (b). The tensile strengths of BFRP and GFRP bars basically decrease with an increase in exposure period for both solutions at all temperatures. The strange phenomena of a sort of "strength resurrection" for NGT40 and NGT55 should be read with caution. This may be caused by the high coefficient of variation. However, the degradations were much accelerated at higher temperatures. As shown in Fig. 4(a), after 63 days of exposure to N-SWSSC solution at 32, 40, 48 and 55 °C, the tensile strength retention of BFRP was 92.7%, 81.7%, 59.1% and 26.0%, respectively. After the 63 days of exposure to HP-SWSSC solution at 32, 40 and 55 °C, the tensile strength retention of BFRP is 97.9%, 90.2% and 77.4%, respectively. Similarly, Fig. 4(b) shows that after 63 days of exposure to N-SWSSC solution at 32, 40 and 55 °C, the tensile strength retention of GFRP was 87.4%, 90.8% and 80.1%, respectively, while after 63 days of exposure to HP-SWSSC solution at 32, 40 and 55 °C, the tensile strength retention of GFRP was 97.9%, 94.1% and 89.6%. These data suggest that N-SWSSC solution caused more damage to BFRP and GFRP bars than HP-SWSSC solution, which may be attributed to the greater alkali-ion content of the N-SWSSC solution than HP-SWSSC solution. Fig. 4(a) and (b) also show the durability of GFRP bars to be much superior than BFRP bars in both N-SWSSC solution and HP-SWSSC solution, especially at 55 °C.

The data for BFRP in the two SWSSC solutions are compared in Fig. 5 with the similar reported data in an alkaline solution [13]. It is clear that N-SWWSC solution is most aggressive to BFRP bars, whereas the HP-SWSSC solution is the least aggressive, and the reported alkaline solution [13] lies in between. This can be explained by the different pH of three solutions. The pH of N-SWSSC, HP-SWSSC and alkaline solutions are 13.4, 12.7 and 12.9–13.1, respectively. Results suggest the higher the pH, the more aggressive the solution is to BFRP bars. Coricciati et al. reported that the crystallographic structure of basalt to consist olivine (single tetrahedron), pyroxenes (linear chain) and plagioclase (tetrahedral space) structures. The reduction in strength is attributed to the breaking of some bonds of the linear tetrahedral chain of pyroxenes during exposure to the aggressive alkaline environment [37].

#### 3.1.2. Stress-strain curves and Young's modulus

As shown in Fig. 6, all the stress-strain curves of BFRP and GFRP bars under different exposure conditions show a linear elastic behaviour until tensile failure. Failures are typically of brittle nature for FRP bars with unidirectional fibres. In addition, the linear region of each curve lies close to each other, which suggest a negligible change in the material's Young's modulus.

Fig. 7 compares the tensile elastic modulus of FRP bars under different exposure conditions. These results indicated that Young's modulus of BFRP and GFRP bars are both not affected by exposure to N-SWSSC and HP-SWSSC solutions. Similar test results for BFRP bars and GFRP bars under various environment conditions are also reported in the literatures [13,14]. This phenomenon is possibly explained by the following theory:

The Young's modulus of unidirectional FRP composites along the reinforced fibre direction is expressed as [38]:

$$E_c = E_m V_m + E_f V_f \tag{1}$$

where $E_c$, $E_m$ and $E_f$ represent the Young's modulus of the FRP composite, resin matrix and reinforced fibre, respectively; and $V_m$ and $V_f$ represent the volume fraction of resin matrix and reinforced fibre inside the FRP composite, respectively.

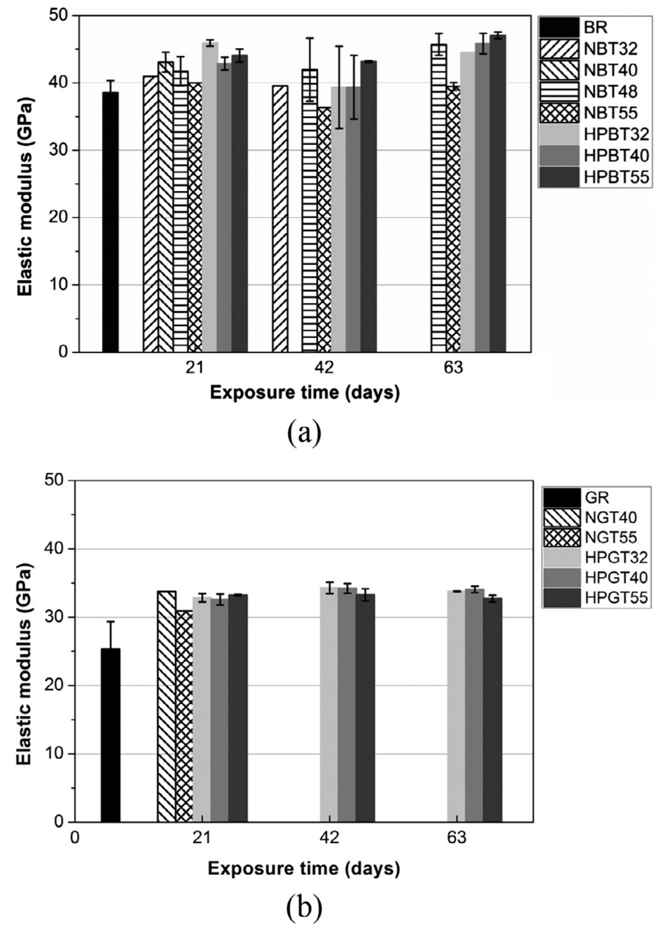



**Fig. 7.** Elastic modulus of conditioned FRP bars in SWSSC solutions: (a) BFRP and (b) GFRP.

Young's modulus of basalt and glass fibres used in this study (93.1–110 GPa and 76 GPa, respectively) are considerably higher than that of epoxy resin (3.6 GPa cited [13]). The modulus of unidirectional FRP bars will mainly depend on the modulus of fibres. Since the degradation of modulus of fibres under SWSSC solutions may not be significant, the Young's moduli of GFRP and BFRP have not suffered any obvious change after exposure.

### 3.2. Failure morphologies

#### 3.2.1. Surface appearance

Fig. 8 shows the surface appearance of BFRP and GFRP specimens that were pre-exposed to N-SWSSC and HP-SWSSC solutions for 63 days at different temperatures. As shown in Fig. 8, the surface colour of BFRP exposure specimens changed obviously (compared with the reference specimen) and the changes in N-SWSSC solution are more intense than those in HP-SWSSC solution, suggesting surface damage of BFRP specimens. More severe surface fuzz and porousness was for the specimens exposed to the N-SWSSC solution at 55 °C condition. It is believed that the degradation of matrix and fibre-resin interface were both significant, which was mainly caused by the N-SWSSC solution attack at the elevated temperatures. Similarly, the surface colour of GFRP specimens also changed (compared with the reference specimen) but the discolouration was much less intense than those of BFRP under the same condition. The degradation suggested by the surface appearance is consistent with the tensile strength results.

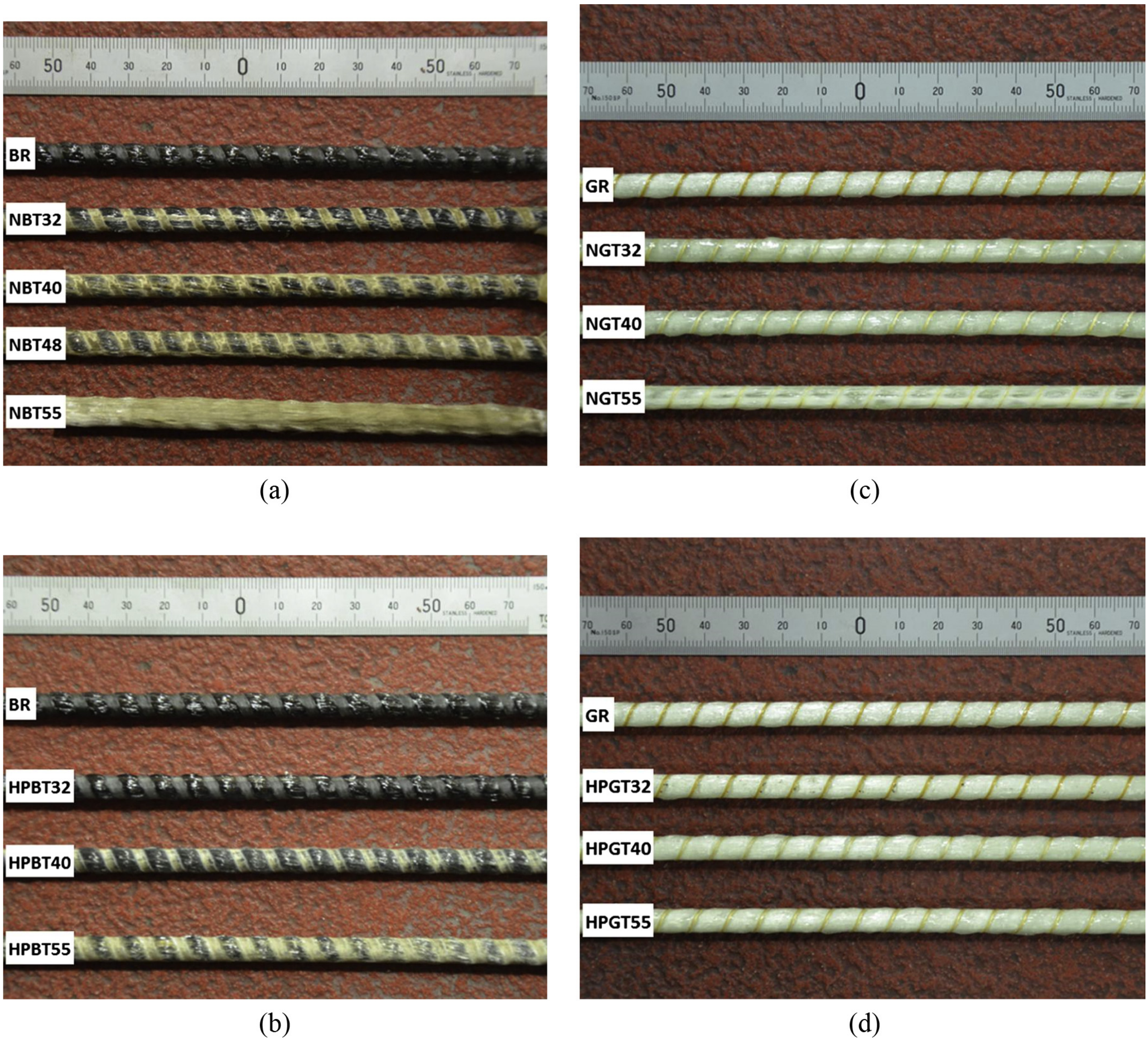



**Fig. 8.** Photographs of conditioned FRP bars after 63 days of exposure: (a) BFRP in N-SWSSC solution; (b) BFRP in HP-SWSSC solution; (c) GFRP in N-SWSSC solution and (d) GFRP in HP-SWSSC solution

### *3.2.2. SEM images*

To explain the damage mechanism of BFRP and GFRP bars in SWSSC solutions, SEM images of the bars after pre-exposure to N-SWSSC solution for 63 days were investigated. Fig. 9 shows the schematics of evolution of the damage along the cross section of FRP bars that have been developed on the basis of the SEM observations. For both GFRP and BFRP bars, the damage starts from the circumferential part as a result of such areas being the first point of contact with the aggressive SWSSC solution. It is clear that the damaged areas increase with the exposure temperature. For BFRP specimen, the damage depths are 100 μm, 300 μm and 2000 μm at 32, 40 and 55 °C, respectively, suggesting the severity of damage increase with the exposure temperature. Under the same exposure condition, the damage of GFRP is much less than that of BFRP. For example, the obvious damage depth for GFRP was limited to ~200 μm at 55 °C, while no obvious damage occurred at 32 and 40 °C condition. This phenomenon proves the superior durability of GFRP than BFRP in SWSSC solution, which is also consistent with the tensile strength results (Fig. 4(a) and (b)).

Fig. 10 shows the SEM images of BFRP bars after pre-exposed to N-SWSSC solution for 63 days. For BFRP reference specimen, the basalt fibre, resin and the fibre-resin interface are all intact, only a few little cracks near the fibre edges are seen, and these were presumably caused by the polishing process during the SEM specimen preparation. For the conditioned specimens, the areas identified as B, C, D and E in Fig. 9(a) suggested that the circumferential parts of BFRP specimens to have suffered damage. The damage mainly includes fibre ruptures, resin degradation and debonding of fibre-resin interface. However, for all specimens, the interior part of BFRP had no obvious damages. Only some damages were found in the core area at 55 °C.

Fig. 11 shows the SEM images of GFRP after pre-exposed to N-SWSSC solution for 63 days. Only for NGT55D63 specimen, the circumferential damages clearly. Like BFRP bar, the damage mainly includes fibre ruptures, resin degradation and debonding of fibre-resin interface. However, for the exposures at 32 and 40 °C, no obvious damages were found. This also proves the durability of GFRP to be much superior than BFRP.

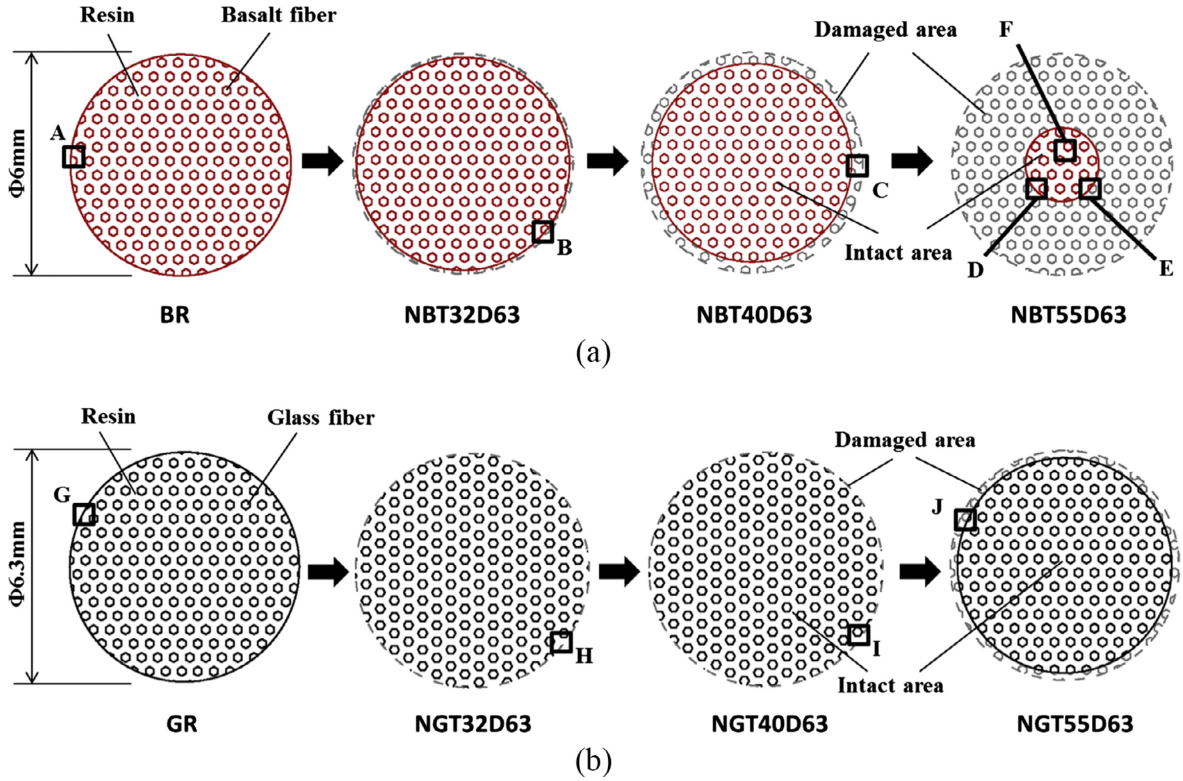



**Fig. 9.** Schematics of evolution of the damage along cross section of the FRP bars after conditioned exposure: (a) BFRP and (b) GFRP.

*3.2.3. Chemical analyses and damage mechanism*

The EDS results for BR, NBT55D63, GR and NGT55D63 at the locations identified by numbers ("1", "2", "3" ...) in Figs. 10 and 11 are summarized in Table 5. The results show that main elements of the reference basalt fibre and E-glass fibre are Si and O where the contents are 27% for Si in both fibres and 44% and 48% for O in basalt and E-glass respectively, because $SiO_2$ is the main chemical constituent of both basalt and glass fibres [39]. The two also contain equal amount of Al element, while the amount of Ca in basalt fibre is lower than that in the E-glass fibre. Na, Mg, K, Fe and Ti were found exclusively in the basalt fibre. The need of these minor elements may make the manufacture of basalt fibre more complex than that of E-glass fibre [40]. The presence Fe existing as $Fe_2O_3$ and FeO in basalt fibre enables the colour distinction of basalt fibre from glass fibre [39–42].

For NBT55D63 specimen, spectrum E-2 in Table 5 shows that the $Cl^-$ content is increased in the resin in the damage area near edge of BFRP bar however its concentration is decreased in the core area as noticed from spectrum F-4. The contents of Na, K, and Ca in the resin generally remain unchanged in the outer damage area. Spectrum E-1 and F-3 show that the content of elements of intact basalt fibre almost remain unchanged except for the Fe which is decreased slightly due to the degradation of basalt fibre as a result of reaction with chloride ions in SWSSC solution [39]. For NGT55D63 specimen, spectrum J-2 and J-4 show the content of Cl, Na and K increases clearly in the resin in the damage edge area of GFRP, while with the distance from the edge increase to more than 200 μm, the contents of Cl, Na and K reduce to the same level of GR specimen. Like BFRP, for spectrum J-1 and J-3 there are no obvious element changes in the intact glass fibre even in the damaged area of GFRP. These results show that $Cl^-$, $Na^+$ and $K^+$ are more easily penetrated into the epoxy of GFRP through the channels. Above test results are consisting with the reported literatures [20,43,44]. The intact outer epoxy resin layer seems to be acting as an effective semipermeable membrane, which only allows water molecules to permit but block $Cl^-$, $Na^+$, $K^+$ and $Ca^{2+}$ [20]. As the alkaline environment only affects the surface layer of glass and basalt fibre, there are no composition changes for intact part of basalt and glass fibres after exposure to alkaline environment [41,43,45].

The damage mechanism of BFRP and GFRP bars in SWSSC solutions includes the degradation of fibres, resin and fibre-resin interface. As reported [12], the degradation of E-glass fibres mainly includes the leaching by the water molecules and the etching by the free hydroxyl ions ($OH^-$). Especially for the etching in alkaline solution, the free hydroxyl ions ($OH^-$) aggressively break the Si-O-Si bone of glass fibre [12,46]. Because the basalt fibre is the closest analog of glass fibre [47], it is possible to apply the etching mechanism of glass fibre to explain the degradation of basalt fibres. In SWSSC solutions, the hydroxyl ions attack the Si–O–Si bone structure of both basalt and glass fibres as shown in the following equation:

$$\left[ -\mathrm{Si{-}O{-}Si}- \right] + OH^- \longrightarrow -\mathrm{Si}-OH\ \text{(solid)} + -\mathrm{Si}-O^-\ \text{(solution)} \qquad (2)$$

The second reaction products ($SiO^-$) of Eq. (2) can further react with water molecules to produce $OH^-$ as following equation:

$$\left[ -\mathrm{Si}{-}\mathrm{O}^- \right] + H_2O \longrightarrow -\mathrm{Si}{-}\mathrm{OH} + OH^- \quad (3)$$

Both fibre degradation processes (Eqs. (2) and (3) above) produce SiOH, which is a gel type product and is less dense than the original fibre structure, and compromises the strength of the fibre-matrix interface. This gel can also transport water and alkalis, which accelerate the fibre degradation [20,48].

In addition, iron that existed only in the basalt fibre can react with the chloride ions from the SWSSC solutions, the main difference in the degradation mechanism between BFRP and GFRP originates from chemical reactions involving $Cl^-$ and $Fe^{2+}$ (that is exclusive to BFRP). When BFRP bars were exposed to SWSSC solutions, $H_2O$, $O_2$ molecules and $Cl^-$ irons can permit into the matrix through the cracks and/or voids and react with the basalt fibre. This caused the colour change of N-SWSSC solution of NBT55D63 specimen after exposure, since iron (II) chloride is known to be yellow. The chemical reactions involved are described as follows [39]:

$$Fe^{2+} + Cl^- \rightarrow [\text{FeCl complex}]^- \quad (4)$$

$$[\text{FeCl complex}]^- + OH^- \rightarrow Fe(OH)_2 + Cl^- \quad (5)$$

$$Fe(OH)_2 + O_2 + H_2O \rightarrow Fe(OH)_3 \rightarrow Fe_2O_3 \cdot nH_2O \quad (6)$$

where, reaction (4) occurs under the condition of [49] $Cl^-/OH^- \geq 0.6$. The final reaction product of rust ($Fe_2O_3{\cdot}nH_2O$) at the

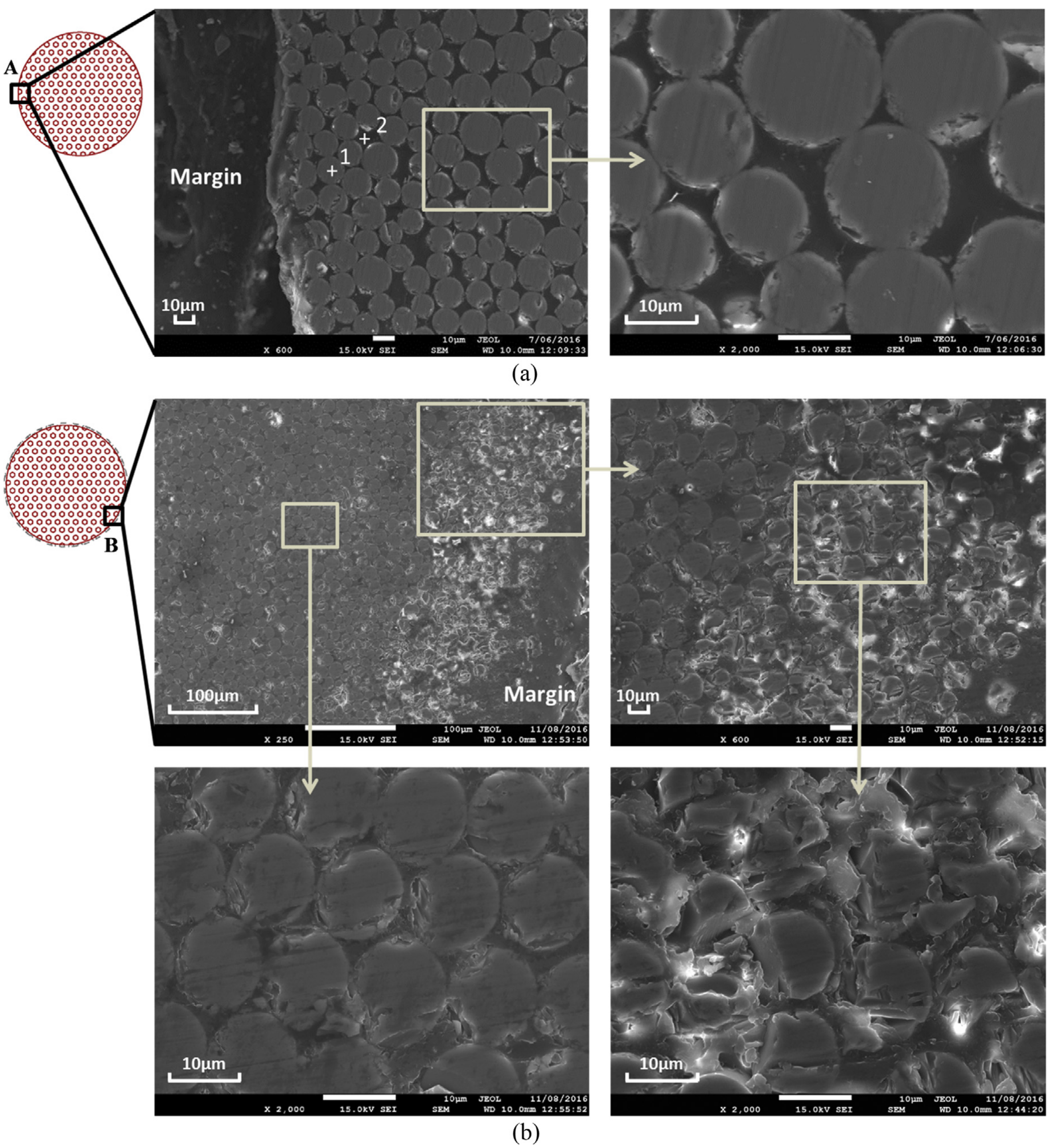
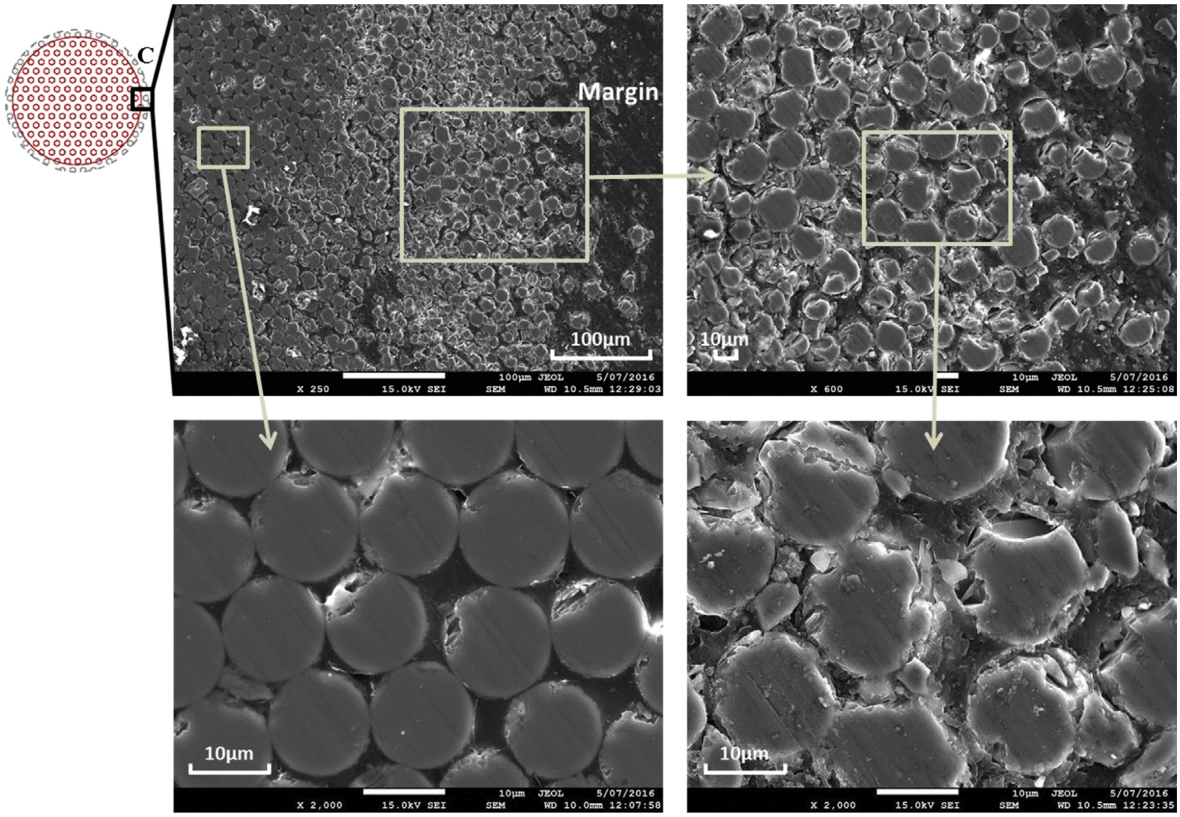
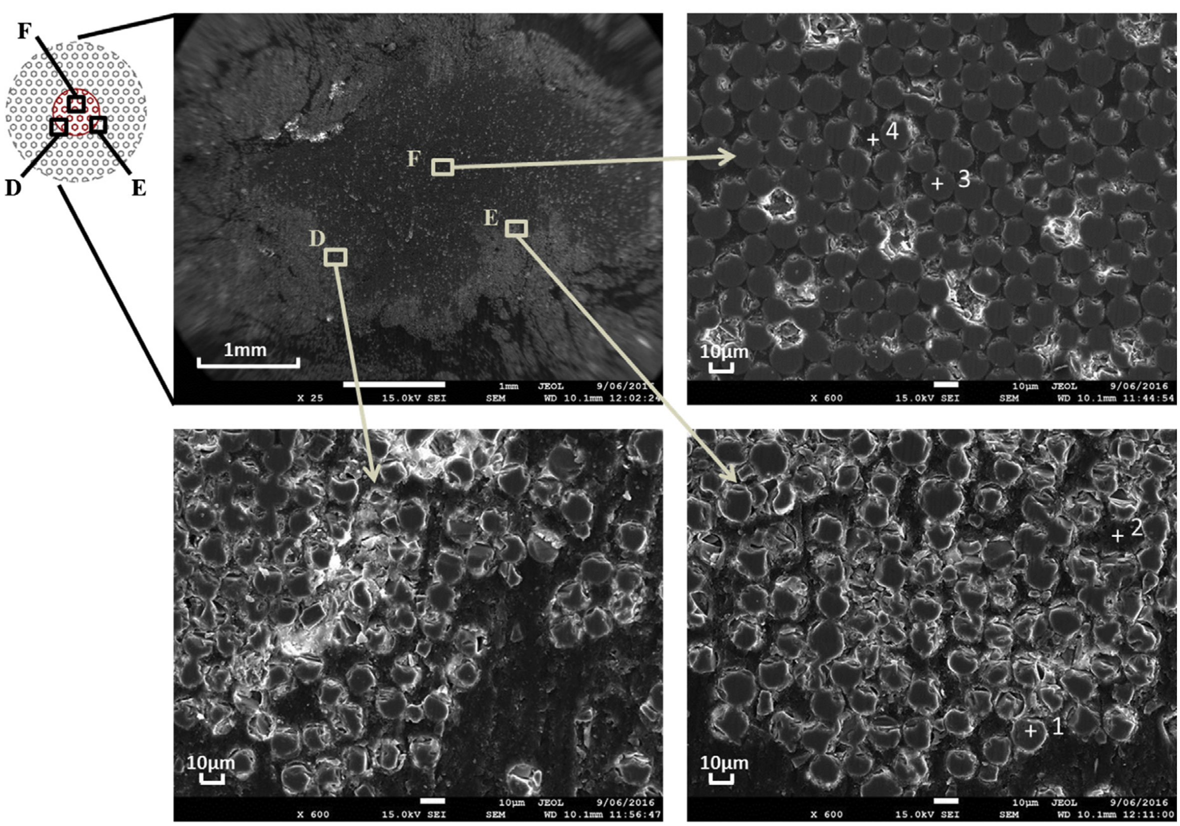



**Fig. 10.** SEM images for BFRP specimens exposed to N-SWSSC solution for 63 days: (a) BR; (b) NBT32D63; (c) NBT40D63 and (d) NBT55D63.




(c)




(d)

**Fig. 10** (*continued*)

interface can absorb more $H_2O$ molecules. This will cause the matrix to expand, making the formation of microcracks easy under a tensile stress. Therefore, this may explain the greater damage of BFRP bars in SWSSC solutions than that of GFRP.

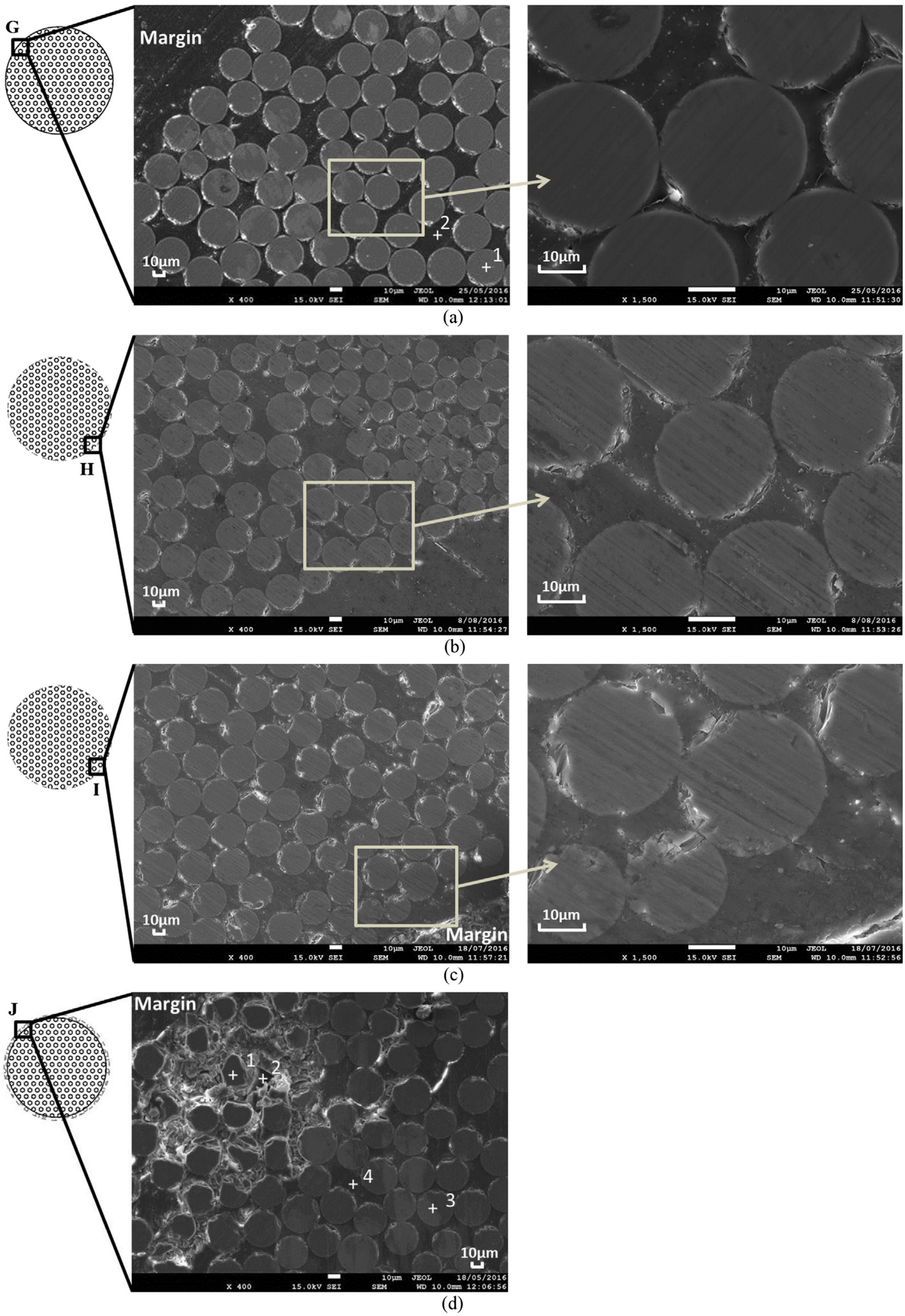



**Fig. 11.** SEM images for GFRP specimens exposed to N-SWSSC solution for 63 days: (a) GR; (b) NGT32D63; (c) NGT40D63 and (d) NGT55D63.

**Table 5**
EDS test results for BFRP and GFRP specimens after exposure to N-SWSSC solution for 63 days.

| Specimen label | Spectrum | Percentage of chemical elements in weight (%) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | O | Na | Mg | Al | Si | Cl | K | Ca | Ti | Fe | Total |
| BR | A-1 | 44 | 3 | 3 | 8 | 27 | – | 2 | 5 | 1 | 7 | 100 |
| BR | A-2 | 81 | – | 1 | 2 | 5 | 6 | – | 2 | – | 3 | 100 |
| NBT55D63 | E-1 | 50 | 3 | 3 | 8 | 25 | – | 2 | 5 | – | 4 | 100 |
| NBT55D63 | E-2 | 81 | – | – | 1 | 3 | 14 | – | – | – | 1 | 100 |
| NBT55D63 | F-3 | 54 | 3 | 3 | 8 | 23 | – | 2 | 4 | – | 3 | 100 |
| NBT55D63 | F-4 | 72 | 2 | 2 | 4 | 14 | 2 | 1 | 2 | – | 1 | 100 |
| GR | G-1 | 48 | – | – | 8 | 27 | – | – | 17 | – | – | 100 |
| GR | G-2 | 65 | – | – | 6 | 19 | 1 | – | 9 | – | – | 100 |
| NGT55D63 | J-1 | 49 | – | – | 8 | 26 | – | – | 17 | – | – | 100 |
| NGT55D63 | J-2 | 47 | 13 | – | 2 | 16 | 13 | 3 | 6 | – | – | 100 |
| NGT55D63 | J-3 | 48 | – | – | 8 | 27 | – | – | 17 | – | – | 100 |
| NGT55D63 | J-4 | 78 | – | – | 3 | 9 | 3 | – | 7 | – | – | 100 |

Note: "–" represents the percentage is less than 0.5%.

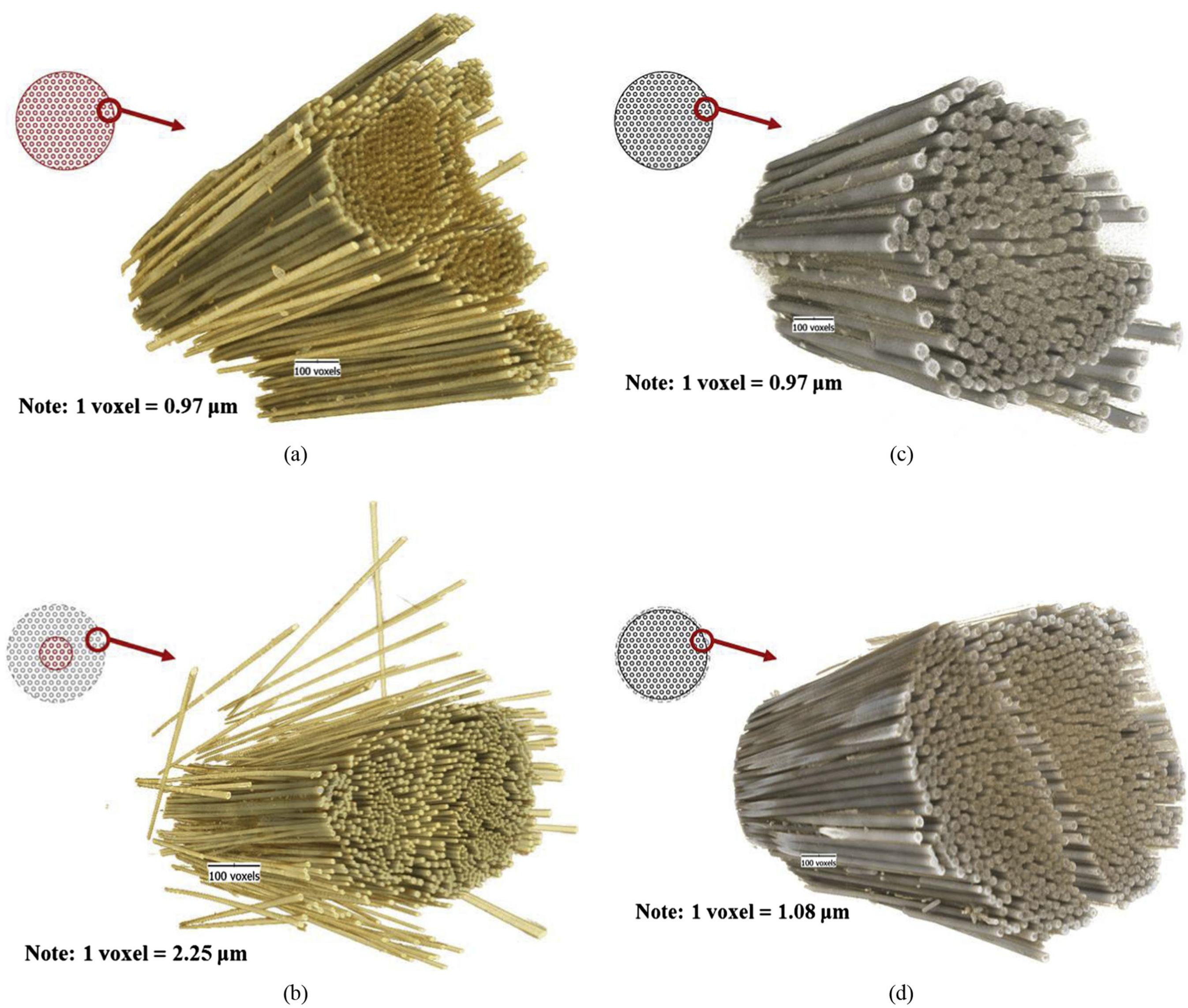
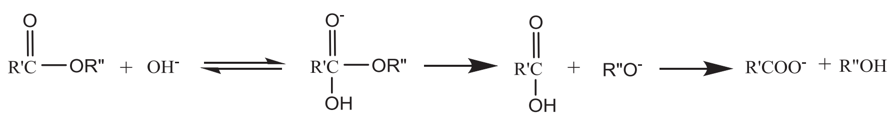



**Fig. 12.** X-ray CT images for distribution of fibres in FRP specimens exposed to N-SWSSC solution for 63 days: (a) BR; (b) NBT55D63; (c) GR and (d) NGT55D63.

The ester is known to be the weakest bond in the polyester and vinyl eater, and the ester hydrolysis is the main factor causing the degradation of matrix [12,20,50]. However, for epoxy resin, the curing agent has a great influence on the molecular structure of the cured resin, whereas when epoxy is mixed with amine curing agent, there are no ester group in the cured resin. When epoxy is mixed with the anhydride curing agent, there are also ester groups in the cured resin. As the curing agents of BFRP and GFRP bars in this study were both the anhydride hardener, the degradation of matrix in BFRP and GFRP is from the degradation of ester by hydrolysis as shown in Eq. (7) [51]:




(7)

Since the fibre-matrix interface is a heterogeneous area between the matrix and the fibres and is ~1 μm thick, the damage at the fibre-matrix interface involves a much more complex mechanism [14]. The damage of fibre-matrix interface for FRP composites is generally in the form of debonding between fibre and resin, and is mainly caused by two parts. The first part is mainly the breakage of chemical bonding between fibre and resin due to the chemical corrosion, and the other part is the weak interlocking between the fibres and resin due to the swelling of resin during the water absorption [52].

### 3.3. X-ray CT image analysis

3-D X-ray CT images for the reference and conditioned BFRP and GFRP bars are shown in Figs. 12-13.

Fig. 12 shows the fibre distribution in BFRP and GFRP specimens before and after pre-exposure to N-SWSSC solution for 63 days at 55 °C. A small volume of the FRP bars were chosen from the outer part of the whole bars (identified by red circles in Fig. 12) for the X-ray CT imaging. For the BFRP reference specimen, the distribution of the fibres is regular and tight (Fig. 12(a)), while the distribution of fibres in NBT55D63 specimen becomes relatively loose and the distance between the neighbourhood fibres increases (Fig. 12(b)). This indicates that the outside part of BFRP bars was damaged severely due to the degradation of resin and the fibre-resin

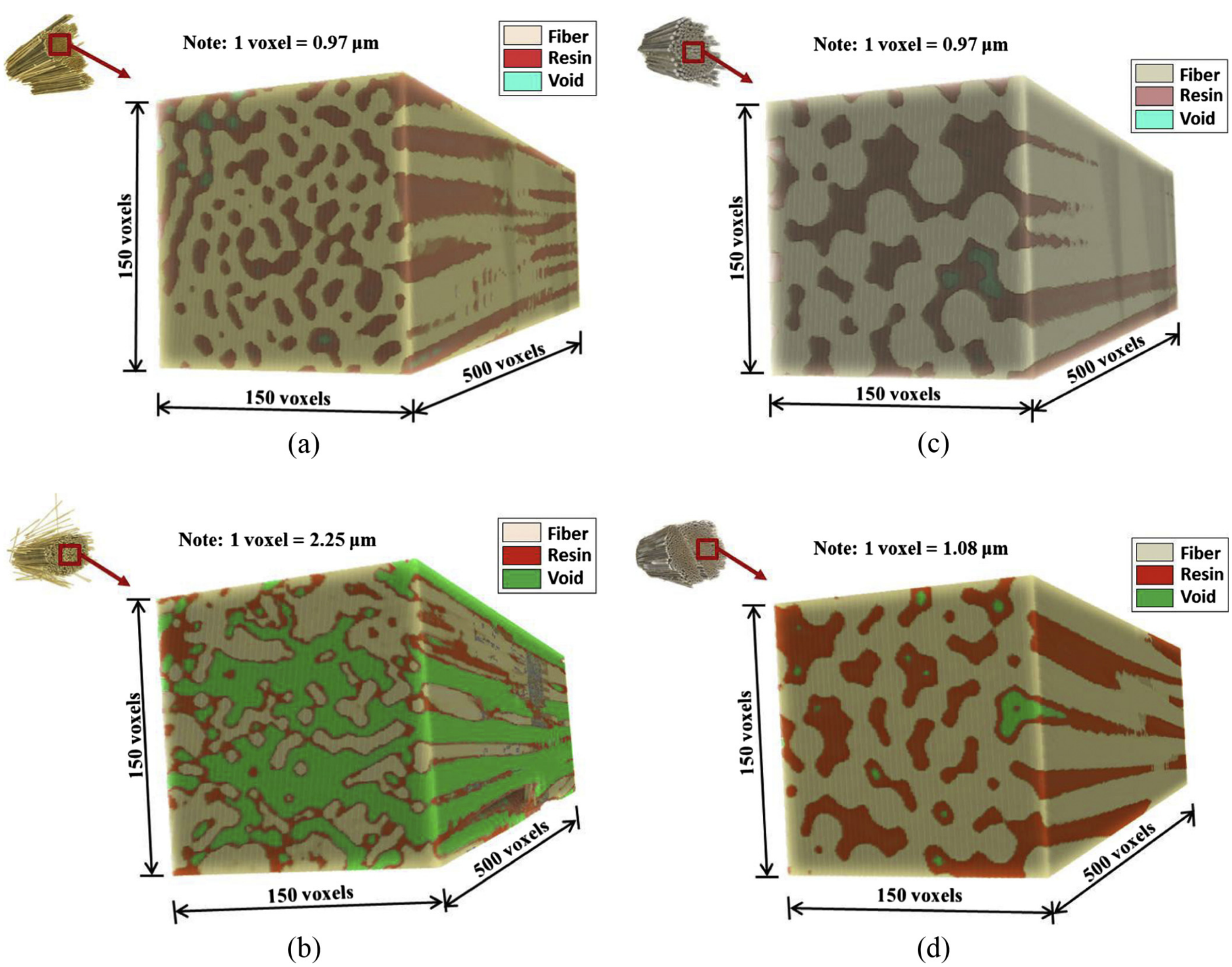



**Fig. 13.** X-ray CT images for FRP specimens exposed to N-SWSSC solution for 63 days: (a) BR; (b) NBT55D63; (c) GR and (d) NGT55D63.

**Table 6**
Volume content of component inside BFRP and GFRP bars calculated from X-ray CT test.

| Specimen label | Volume fraction (%) Fibre | Resin | Void |
|---|---|---|---|
| BR | 63.7 | 35.7 | 0.6 |
| NBT55D63 | 41.1 | 17.8 | 41.1 |
| GR | 66.7 | 33.0 | 0.3 |
| NGT55D63 | 65.7 | 32.9 | 1.4 |

interfaces. However, for the GFRP, after pre-exposure to N-SWSSC solution for 63 days at 55 °C, the distribution of fibres in the outside part of the specimen did not show any obvious changes (Fig. 12(c) and (d)). Only one crack caused during the specimen preparation for the NGT55D63 specimen (refer to Fig. 12(d)) can be ignored for the qualitative assessment of the performance of bars. This observation implies a relatively insignificant damage of the GFRP bars in comparison with the BFRP bars.

The volumes of fibres, resin and voids present in the BFRP and GFRP specimens were calculated by selecting a small representative elementary volume (REV) of size 150 × 150 × 500 voxels (Fig. 13). It is clear that for the BFRP specimen, the void volume increased and both the fibre and resin volumes decreased with exposure. But for the GFRP specimen, an insignificant change of resin-fibre-void phase was observed. The volumes of all phases for the FRP bars before and after exposure are summarized in Table 6. The results show that for the BFRP, the content of the fibre and resin reduced considerably (from 63.7% to 41.1% for fibre; from 35.7% to 17.8% for resin). On the other hand, the void content for the BFRP increases dramatically from 0.6% to 41.1% after exposure, while for the GFRP, the content of void increases marginally from 0.3% to 1.4%. The volume contents of the fibre and the resin for the GFRP remained unchanged after exposure. This again confirms the durability of GFRP to be superior than that of the BFRP, which is also in-line with the results of the SEM analysis.

## 4. Prediction of long-term behaviour of FRP bars

### 4.1. Three existing models

To evaluate the long-term performance of FRP bars in civil engineering environment, the popular Arrhenius relation was adopted by researchers based on the short-term data from accelerated aging tests [53]. The basic assumption of Arrhenius relation is that the single dominant degradation mechanism of the material does not change with time and temperature during the exposure, whereas the rate of degradation is accelerated with the increase in temperature. Based on the Arrhenius relation, several prediction models were proposed by researchers [8,13,16,18,54–58]. Among these models, three models were most frequently used by the researchers.

Model 1 was proposed by Bank et al. [55], which suggested a service life prediction procedure for FRP materials that the accelerated aging data could be calculated and plotted with the property retention in linear scale versus time in logarithmic scale. Though this model has been successfully used to predict the long-term tensile strength of GFRP bars embedded in the concrete [14,16], there are still some intrinsic limitations in this model [8]. On the one hand, this model is only a phenomenological representation of test data and never provides any hypothesis of the degradation mechanism. On the other hand, according to this model, the residual strength approaches infinity at aging time zero, which is obviously contradictory with the real test data. In addition, Wu et al. [13] found that the Arrhenius plots fitted using the tested data of BFRP bars were not parallel to each other, which indicated that the degradation mechanism maybe changed with exposure time, violating the fundamental Arrhenius assumption [8].

Different from Model 1, Model 2 and Model 3 both consider that the degradation rate of mechanical properties of FRP composites is high in the initial stage, and then reduces with the increase of exposure time. They were firstly proposed by Phani and Bose with the acousto-ultrasonic technique to predict flexural strength retention of composite laminates [56,57], and then modified to predict the tensile strength of GFRP and BFRP bars in alkaline concrete pore environment [8,13,18]. The degradation mechanism of FRP composites for these two models is assumed to be debonding at the fibre/matrix interface, which was also confirmed with the

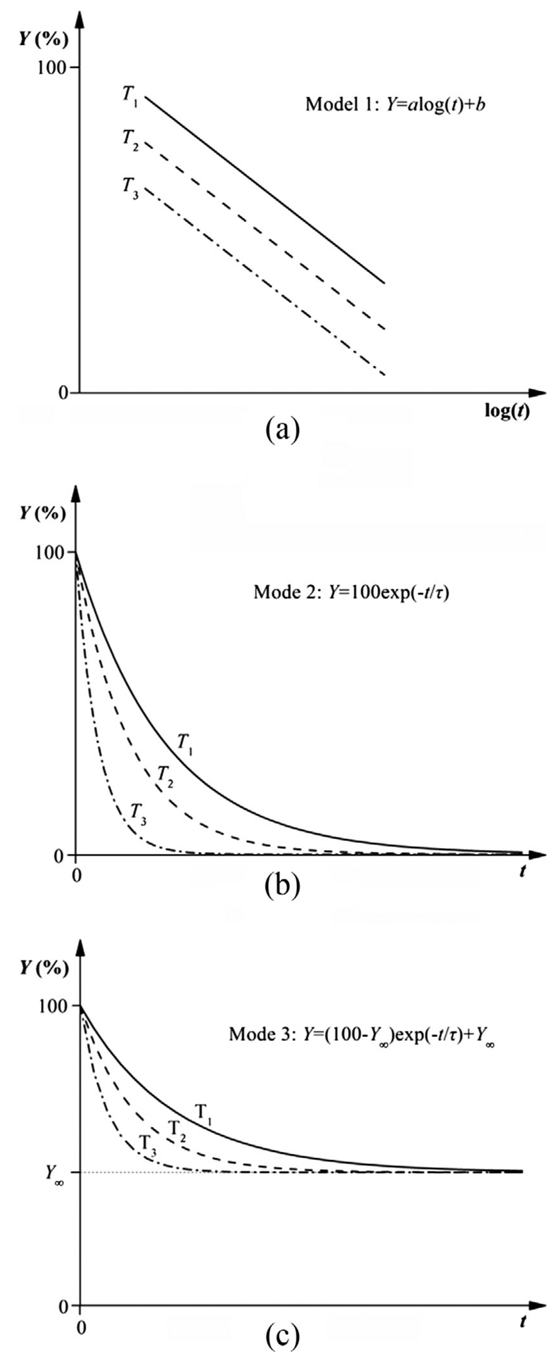
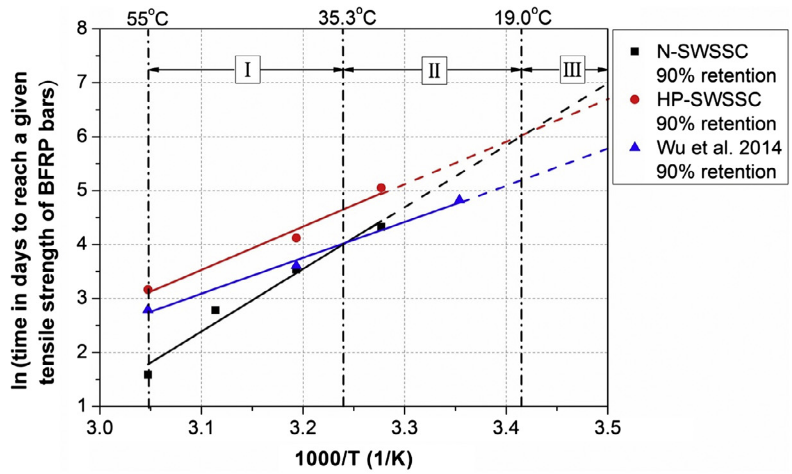
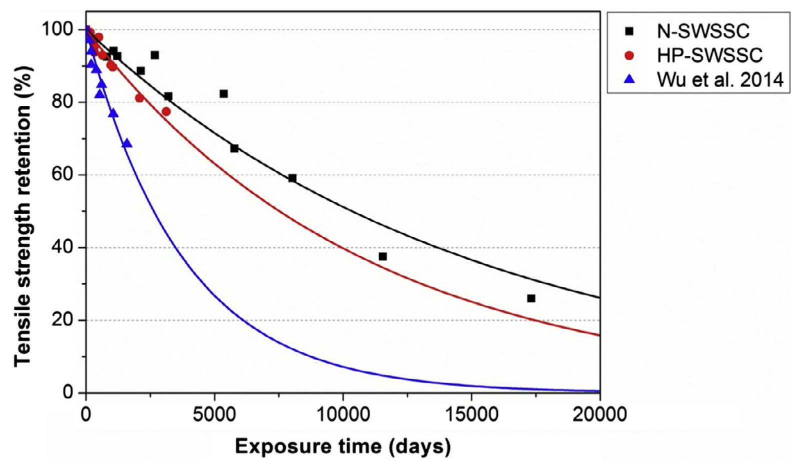



**Fig. 14.** Schematics of three models used for long-term life prediction of FRP materials: (a) Model 1; (b) Model 2 and (c) Model 3.



(a)



(b)

**Fig. 15.** Comparison of predictions for BFRP bars at 9.9 °C using Model 2: (a) Arrhenius plots of tensile strength retention and (b) master curves.

**Table 7**
Coefficients of regression equations for Arrhenius plots of BFRP bars in Model 2.

| Solution type | $E_a/R$ | $R^2$ |
|---|---|---|
| N-SWSSC | 11,567 | 0.96 |
| HP-SWSSC | 8039 | 0.96 |
| Alkaline (Wu et al. 2014) | 6646 | 0.99 |

scanning electron microscopy (SEM) images by some researchers [13,20].

Above three existing models are schematized in Fig. 14, and defined by:

$$\text{Model 1}: \quad Y = a\log(t) + b \tag{8}$$

$$\text{Model 2}: \quad Y = 100\exp(-t/\tau) \tag{9}$$

$$\text{Model 3}: \quad Y = (100 - Y_\infty)\exp(-t/\tau) + Y_\infty \tag{10}$$

where $Y$ represents the tensile strength retention (%); $t$ represents the exposure time; $a$ and $b$ represent the regression constants; $\tau$ represents the fitted parameter; and $Y_\infty$ represents the tensile strength retention (%) at exposure time of infinity.

As shown in Fig. 14, It is clear that Model 2 [13,18] and Model 3 [8,56] are similar, but the tensile strength retention reaches zero or a given retention $Y_\infty$ at the exposure time of infinity. In these three models, the degradation rate increases with the increased exposure temperatures ($T_1 < T_2 < T_3$).

In all above three models, the degrade rate can be expressed in the Arrhenius relationship by the following equation [13]:

$$k = A\exp(-E_a/RT) \tag{11}$$

where $k$ represents the degradation rate (1/time); $A$ represents the constant of the material and degradation process; $E_a$ represents the activation energy; $R$ represents the universal gas constant; and $T$ represents the Kelvin temperature. Furthermore, Eq. (11) can be transformed into:

$$\frac{1}{k} = \frac{1}{A}\exp(E_a/RT) \tag{12}$$

$$\ln\left(\frac{1}{k}\right) = \frac{E_a}{R}\frac{1}{T} - \ln A \tag{13}$$

Eq. (12) shows the degradation rate $k$ can be expressed as the inverse of the time needed for a material property to reach a given value. Eq. (13) shows the logarithm of time needed for a material property to reach a given value is a linear function of $1/T$ with the slope of $E_a/R$ value.

According to Eq. (11), the time-shift factor (TSF) for the tensile strength to reach the same value (represented by c) at

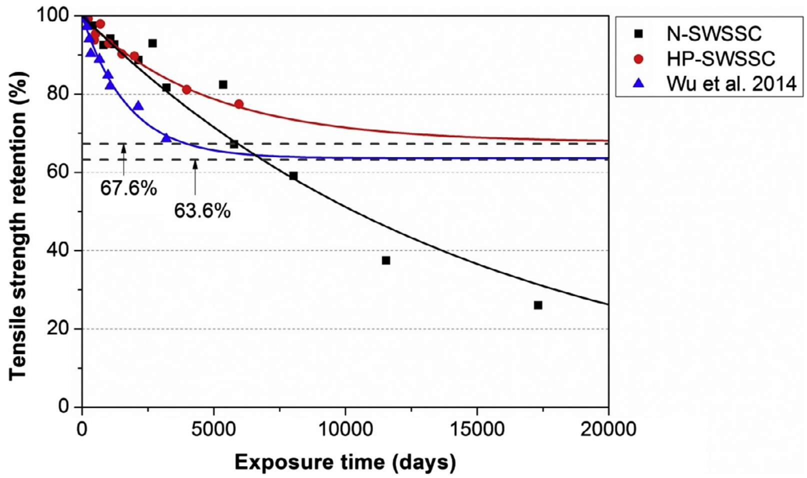



**Fig. 16.** Comparison of predictions for BFRP bars at 9.9 °C using Model 3.

**Table 8**
Long-term predication results of FRP bars at different mean annual temperatures for tensile design using Model 2.

| Structure | Mean annual temperature (°C) | Time in years to reach 70% tensile retention of FRP bars | | | |
|---|---|---|---|---|---|
| | | BFRP in N-SWSSC solution | BFRP in HP-SWSSC solution | BFRP in alkaline solution (Wu et al., 2014) | GFRP embedded in concrete and immersed in the saline solution (Robert et al., 2013) |
| Hall's Harbor Wharf | 7.6[a] | 20.4 | 13.4 | 4.5 | 6.5 |
| Joffre Bridge | 4.1 | 34.3 | 19.2 | 6.1 | 7.0 |
| Chatham Bridge | 4.6 | 31.8 | 18.2 | 5.8 | 6.9 |
| Crowchild Trail Bridge | 3.9 | 35.3 | 19.6 | 6.2 | 7.0 |
| Waterloo Creek Bridge | 9.9 | 14.6 | 10.6 | 3.7 | 6.2 |

Note: a: annual mean temperature from the World Climate & Temperature online database by 12 August 2016. (http://www.climatemps.com/).

**Table 9**
Long-term predication results of FRP bars at different mean annual temperatures using Model 3.

| Structure | Mean annual temperature (°C) | Time in years to reach 70% tensile retention of FRP bars | | | |
|---|---|---|---|---|---|
| | | BFRP in N-SWSSC solution | BFRP in HP-SWSSC solution | BFRP in alkaline solution (Wu et al., 2014) | GFRP embedded in concrete and immersed in the saline solution (Robert et al., 2013) |
| Hall's Harbor Wharf | 7.6[a] | 20.4 | 43.8 | 10.4 | 18.8 |
| Joffre Bridge | 4.1 | 34.3 | 66.8 | 15.0 | 20.3 |
| Chatham Bridge | 4.6 | 31.8 | 62.9 | 14.2 | 20.1 |
| Crowchild Trail Bridge | 3.9 | 35.3 | 68.5 | 15.3 | 20.4 |
| Waterloo Creek Bridge | 9.9 | 14.6 | 33.4 | 8.3 | 17.8 |

Note: a: annual mean temperature from the World Climate & Temperature online database by 12 August 2016. (http://www.climatemps.com/).

temperatures $T_1$ and $T_0$ can be obtained from the previous Arrhenius plots. The TSF can be expressed as:

$$\mathrm{TSF} = \frac{t_0}{t_1} = \frac{c/k_0}{c/k_1} = \frac{k_1}{k_0} = \frac{A\exp(-E_a/RT_1)}{A\exp(-E_a/RT_0)} = \exp\left[\frac{E_a}{R}\left(\frac{1}{T_0} - \frac{1}{T_1}\right)\right] \quad (14)$$

Through above three models, the long-term tensile behaviour of FRP bars can be predicted at the any given temperatures, respectively.

There is a need to compare the long-term prediction results using short-term test data with the field results. However, only limited field test data of FRP bars can be found. Mufti et al. investigated five serviced structures in Canada [29]. They found that the GFRP bars after serviced in concrete structures for 5~8 years were still intact through SEM test. Furthermore the Fourier transform infrared (FTIR) spectroscopy and glass transition temperature (Tg) tests also showed that there are no any chemical elements changes for serviced GFRP bars. To check the reliability of prediction models, the prediction results calculated from the accelerate data are also compared in this paper with the field data from five Canada structures mentioned in [29].

### 4.2. Prediction results

In this part, long-term prediction of BFRP bars in three kinds of solutions (i. e. N-SWSSC, HP-SWSSC and alkaline) were calculated using accelerated test results with Model 2 and Model 3, respectively. No calculation using Model 1 is carried out in this paper due to the issues mentioned in [13].

In addition, since the tested data of GFRP bars generated in this study are not sufficient to develop the prediction using Model 2 and Model 3, the long-term prediction of GFRP bars was calculated using the published data [14].

The detailed prediction of FRP bars using Model 2 and Model 3 are shown in the Appendix A (Fig. A1 and Tables A1–A4) for BFRP and GFRP bars.

The mean annual temperatures of five Canada structures are 7.6 °C for Hall's Harbor Wharf, 4.1 °C for Joffre Bridge, 4.6 °C for Chatham Bridge, 3.9 °C for Crowchild Trail Bridge and 9.9 °C for Waterloo Creek Bridge, respectively. Therefore, the above five annual temperatures were chosen as the prediction temperatures for BFRP and GFRP specimens.

The prediction results of BFRP bars at 9.9 °C using Model 2 are shown in Fig. 15. As shown in Fig. 15(a), the prediction results were divided into three stages according to the exposure temperatures range. If the exposure temperature is among the stage I, the degradation rate of BFRP is highest in the N-SWSSC solution, middle in the alkaline solution [13] and lowest in the HP-SWSSC solution. If it is among the stage II, the degradation rate of BFRP is highest in the alkaline solution [13], middle in the N-SWSSC solution, and lowest in the HP-SWSSC solution. If it is among the stage III, the degradation rate of BFRP is highest in the alkaline solution [13], middle in the HP-SWSSC solution, and lowest in the N-SWSSC solution. For example, the mean annual temperature of Waterloo Creek Bridge is 9.9 °C, which is located in stage III, therefore, the prediction results of BFRP bars at 9.9 °C are shown in Fig. 15(b). It is clear that the tensile strength retentions of BFRP in three kinds of solutions all reach to zero after the indefinite time. The degradation rate of BFRP is the highest in alkaline solution [13] and middle in HP-SWSSC solution, while lowest in N-SWSSC solution. This result looks inconsistent with the accelerate test results, which shows that the degradation rate of BFRP bars in N-SWSSC solution is the highest. This proves that the prediction result of Model 2 is very sensitive to the prediction temperature ranges. This strange phenomenon is caused by the different coefficients of regression equations for Arrhenius plots of BFRP bars in three solutions in Model 2. As summarized in Table 7, the slope ($E_a/R$) of Arrhenius plots for BFRP bars is 11,567 in N-SWSSC, 8039 in HP-SWSSC, and 6646 in alkaline solution, respectively. As pointed by Chen et al., different $E_a/R$ values resulted in different degradation rates and probably mechanisms of FRP bars [18]. Meanwhile, it is noted that the adopted room temperature for N-SWSSC solution is 32 °C, which is different from 25 °C for alkaline solution adopted by [13]. This may cause some calculation error, which will be amplified when linearly extend the Arrhenius plots from accelerated temperature to predicted temperature. Therefore, a larger range of accelerated exposure temperatures are suggested to improve the prediction precision of Model 2.

The prediction results of BFRP bars at 9.9 °C using Model 3 are shown in Fig. 16. The results of Model 3 reveal that in any prediction temperatures, the tensile strength retention of BFRP converges to 67.6% in the HP-SWSSC solution, to 63.6% in the alkaline solution [13], while to 0% in the N-SWSSC solution. This result is basically consistent with the accelerate test results. It is clear that the prediction result of Model 3 is more reasonable than that of Model 2. It is interesting that for BFRP bars in N-SWSSC solution, the results of Mode 3 and Model 2 are the same.

On the other hand, the design tensile strength of FRP bars should be considered when FRP bars are used in concrete. According to ACI 440.1R-06, the environmental reduction factor $C_E$ for GFRP bars in concrete exposed to earth and weather is 70%, while the $C_E$ for BFRP bars is not supplied. Therefore, in this study, the long-term life of BFRP and GFRP bars in five different locations in Canada were also predicted using Model 2 and Model 3, and the time to reach 70% tensile strength retention is summarized in Table 8 and Table 9. As shown in Table 8, according to the prediction result of Model 2, when the tensile strength retention of BFRP reaches to 70%, it will take 14.6 to 35.3 years in N-SWSSC solution, 10.6 to 19.6 years in HP-SWSSC, and 3.7 to 6.2 years in alkaline solution [13]. When the tensile strength retention of GFRP embedded in concrete and immersed in the saline solution reaches to 70%, it will take 6.2 to 7.0 years. As shown in Table 9, according to the prediction result of Model 3, when the tensile strength retention of BFRP reaches to 70%, it will take 14.6 to 35.3 years in N-SWSSC solution, 33.4 to 68.5 years in HP-SWSSC, and 8.3 to 15.3 years in alkaline solution [13]. When the tensile strength retention of GFRP embedded in concrete and immersed in the sal-

**Table 10**
Comparison between long-term prediction results and field results of FRP bars using Model 2.

| Structure | Serviced years by 2007 | Mean annual temperature (°C) | Calculated tensile strength retention of FRP bars after the given serviced years (%) | | | | Retention of field GFRP (%) (Mufti et al., 2007) |
|---|---|---|---|---|---|---|---|
| | | | BFRP in N-SWSSC solution | BFRP in HP-SWSSC solution | BFRP in alkaline solution (Wu et al., 2014) | GFRP embedded in concrete and immersed in the saline solution (Robert et al., 2013) | |
| Hall's Harbor Wharf | 5 | 7.6[a] | 91.6 | 87.5 | 67.3 | 75.9 | 100 |
| Joffre Bridge | 7 | 4.1 | 93.0 | 87.8 | 66.3 | 69.8 | 100 |
| Chatham Bridge | 8 | 4.6 | 91.4 | 85.5 | 61.2 | 66.1 | 100 |
| Crowchild Trail Bridge | 8 | 3.9 | 92.2 | 86.5 | 63.0 | 66.5 | 100 |
| Waterloo Creek Bridge | 6 | 9.9 | 86.4 | 81.7 | 56.2 | 70.7 | 100 |

Note: a: annual mean temperature from the World Climate & Temperature online database by 12 August 2016. (http://www.climatemps.com/).

**Table 11**
Comparison between long-term prediction results and field results of FRP bars using Model 3.

| Structure | Serviced years by 2007 | Mean annual temperature (°C) | Calculated tensile strength retention of FRP bars after the given serviced years (%) | | | | Retention of field GFRP (%) (Mufti et al., 2007) |
|---|---|---|---|---|---|---|---|
| | | | BFRP in N-SWSSC solution | BFRP in HP-SWSSC solution | BFRP in alkaline solution (Wu et al., 2014) | GFRP embedded in concrete and immersed in the saline solution (Robert et al., 2013) | |
| Hall's Harbor Wharf | 5 | 7.6[a] | 91.6 | 91.6 | 79.4 | 81.4 | 100 |
| Joffre Bridge | 7 | 4.1 | 93.0 | 92.2 | 79.8 | 78.5 | 100 |
| Chatham Bridge | 8 | 4.6 | 91.4 | 90.8 | 77.3 | 76.9 | 100 |
| Crowchild Trail Bridge | 8 | 3.9 | 92.2 | 91.4 | 78.3 | 77.1 | 100 |
| Waterloo Creek Bridge | 6 | 9.9 | 86.4 | 87.8 | 73.9 | 78.7 | 100 |

Note: a: annual mean temperature from the World Climate & Temperature online database by 12 August 2016. (http://www.climatemps.com/).

ine solution reaches to 70%, it will take 17.8 to 20.4 years. It is clear that the prediction result of Model 2 is much more conservative than that of Model 3.

Table 10 and Table 11 also summarize the comparison between long-term prediction results and field data of FRP bars using Model 2 and Model 3. As the field test result showed that there are no changes in the microstructures of GFRP bars after 5∼8 years' service in field concrete structures, the tensile strength retentions of GFRP bars in five field structures were considered to keep 100% level after 5∼8 years in this study. It seems that the existing models using accelerated test results may be too conservative. There are two possible reasons: (1) the humidity of the concrete in service condition is obvious lower than that of simulated solution [59], (2) the temperature inside the concrete may be much less than the environmental temperature, and is not constant as used in the prediction model. More research is needed to clarify this issue. In future, a more precise degradation model should be developed for BFRP and GFRP bars in concrete, considering the real temperature ranges, humility ranges as well as pre-stress caused by service load.

## 5. Conclusions

The following observations and conclusions can be made based on limited test data obtained in this paper:

1) In this study, GFRP bars had much better durability than BFRP bars especially at high temperatures. This was mainly due to the resin property of GFRP bar is more stable in SWSSC solutions than that of BFRP bar, as evidenced by SEM and 3D X-ray CT results.
2) Nearly no change was found in Young's Modulus for GFRP and BFRP bars after exposure in SWSSC solutions. This is because that the modulus of BFRP and GFRP bars depend on the Young's Modulus of basalt and glass fibre, while the degradation of modulus of basalt fibre under SWSSC solutions may not be significant.
3) For GFRP and BFRP bars, N-SWSSC solution is more aggressive than HP-SWSSC solution due to the higher alkali ions concentration in N-SWSSC solution.
4) For GFRP and BFRP bars, $Cl^-$ easily permeated into the epoxy resin through the channels or crack/void than other alkali metal ions ($Na^+$, $K^+$ and $Ca^{2+}$). No irons obviously permeated into the basalt fibre or glass fibre.
5) The degradations of GFRP and BFRP bars in N-SWSSC solution at high temperature both mainly include etching of fibre, hydrolysis of resin, and interface debonding. However, the reaction between $Cl^-$ and $Fe^{2+}$ is the inherent degradation for BFRP bars, which probably causes more severe damage of BFRP bars than GFRP bars in SWSSC solution.
6) Arrhenius relationship theory was adopted to predict the long-term performance of BFRP and GFRP bars in five places in Canada. However, the prediction of Model 2 and Model 3 are both conservative.
7) In future, a more precise degradation model should be developed for BFRP and GFRP bars in SWSSC by considering the real temperature ranges, humility ranges and pre-loading stress.

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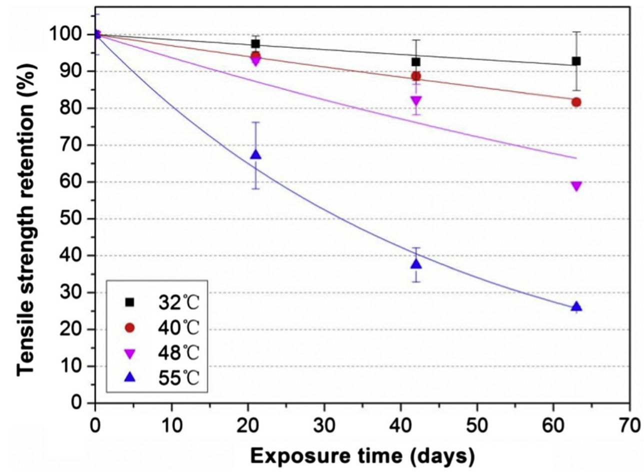
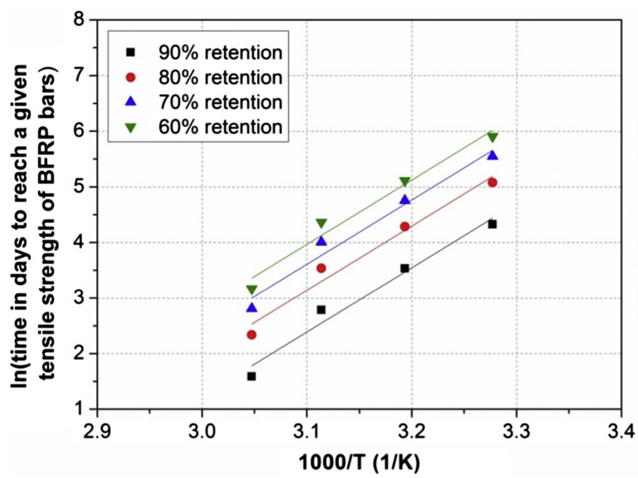
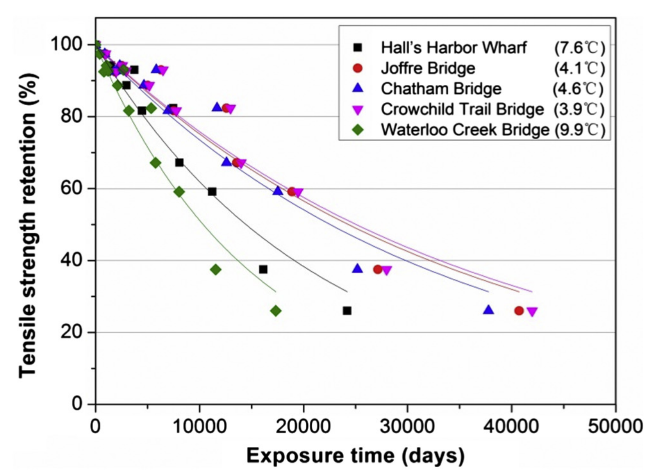


(a)



(b)



(c)

**Fig. A1.** Prediction of BFRP bars in N-SWSSC solution using Model 2 or Model 3: (a) fitted curves; (b) Arrhenius plots of tensile strength retention and (c) master curves.

**Table A1**
Coefficients of the regression equations in Eqs. (9) and (10) for BFRP bars in N-SWSSC solution.

| Temperature (°C) | Model 2 (fitted parameters in Eq. (9)) | | Model 3 (fitted parameters in Eq. (10)) | | |
|---|---|---|---|---|---|
| | $\tau$ | $R^2$ | $\tau$ | $Y_\infty$ | $R^2$ |
| 32 | 719.2 | 0.88 | 719.2 | 0 | 0.88 |
| 40 | 325.8 | 0.99 | 325.8 | 0 | 0.99 |
| 48 | 154.0 | 0.87 | 154.0 | 0 | 0.87 |
| 55 | 46.5 | 0.99 | 46.5 | 0 | 0.99 |

**Table A2**
Coefficients of the regression equations in Eq. (13) for BFRP bars in N-SWSSC solution.

| Tensile strength retention (%) | Model 2 | | Model 3 | |
|---|---|---|---|---|
| | $E_a/R$ | $R^2$ | $E_a/R$ | $R^2$ |
| 60 | 11,567 | 0.96 | 11,567 | 0.96 |
| 70 | 11,567 | 0.96 | 11,567 | 0.96 |
| 80 | 11,567 | 0.96 | 11,567 | 0.96 |
| 90 | 11,567 | 0.96 | 11,567 | 0.96 |

**Table A3**
Time-shift factor for BFRP bars in N-SWSSC solution.

| Model | Temperature (°C) | Time-shift factor (TSF) | | | | |
|---|---|---|---|---|---|---|
| | | Hall's Harbor Wharf 7.6 °C | Joffre Bridge 4.1 °C | Chatham Bridge 4.6 °C | Crowchild Trail Bridge 3.9 °C | Waterloo Creek Bridge 9.9 °C |
| Model 2 | 32 | 27.0 | 45.4 | 42.1 | 46.7 | 19.3 |
| | 40 | 71.0 | 119.5 | 110.8 | 123.1 | 50.8 |
| | 48 | 178.2 | 299.8 | 278.1 | 308.9 | 127.5 |
| | 55 | 384.2 | 646.4 | 599.6 | 666.1 | 274.9 |
| Model 3 | 32 | 27.0 | 45.4 | 42.1 | 46.7 | 19.3 |
| | 40 | 71.0 | 119.5 | 110.8 | 123.1 | 50.8 |
| | 48 | 178.2 | 299.8 | 278.1 | 308.9 | 127.5 |
| | 55 | 384.2 | 646.4 | 599.6 | 666.1 | 274.9 |

**Table A4**
Coefficients of the regression equations for the master curves of BFRP bars in N-SWSSC solution.

| Structure | Mean annual temperature (°C) | Model 2 | | Model 3 | | |
|---|---|---|---|---|---|---|
| | | $\tau$ | $R^2$ | $\tau$ | $Y_\infty$ | $R^2$ |
| Hall's Harbor Wharf | 7.6 | 20865.5 | 0.95 | 20865.5 | 0 | 0.95 |
| Joffre Bridge | 4.1 | 35100.3 | 0.95 | 35100.3 | 0 | 0.95 |
| Chatham Bridge | 4.6 | 32560.7 | 0.95 | 32560.7 | 0 | 0.95 |
| Crowchild Trail Bridge | 3.9 | 36173.5 | 0.95 | 36173.5 | 0 | 0.95 |
| Waterloo Creek Bridge | 9.9 | 14929.1 | 0.95 | 14929.1 | 0 | 0.95 |

## Appendix A. Procedure for the long-term prediction of tensile strength for BFRP and GFRP bars

As an example, Fig. A1 and Tables A1–A4 show the detailed prediction procedure of BFRP bars in N-SWSSC solution using Model 2 and Model 3.

The prediction procedures of BFRP bars in different solutions using Model 2 are identical. In the case of BFRP bars in N-SWSSC solution, the calculation procedure is as following:

Firstly, on the basis of Eq. (9), the fitted curves are shown in Fig. A1(a), and corresponding $\tau$ values and correlation coefficients ($R^2$) are listed in Table A1. The $R^2$ of all the curves are above 0.87.

Secondly, with the regression coefficient $\tau$ listed in Table A1, the time ($t$) needed for the tensile strength retention to reach 60, 70, 80, and 90% at 32, 40, 48 and 55 °C were calculated from Eq. (9). Next, Eq. (13) was fitted to those values ($\ln(1/k) = \ln t$ and $1/T$), and the straight lines were exactly parallel. The slopes of the straight lines ($E_a/R$) and the correlation coefficients ($R^2$) are listed in Table A2.

Thirdly, according to Eq. (14), the TSF values with reference temperatures $T_0$ equal to annual mean temperature of each chosen city were calculated and listed in Table A3.

Finally, once the TSF values for 32, 40, 48 and 55 °C were obtained, Fig. A1(a) was transformed into Fig. A1(c) by multiplying exposure times at 32, 40, 48 and 55 °C with corresponding TSF values. The master curves for tensile strength retention versus exposure time at five different Canada locations were obtained by fitting Eq. (9) in Fig. A1(c), and fitted $\tau$ values and correlation coefficients ($R^2$) are listed in Table A4.

Similarly, the prediction procedures of BFRP bars in different solutions using Model 3 are also identical. In addition, the calculation procedure of Model 3 is same to Model 2, only except for replacing Eq. (9) with Eq. (10). It is noted that the predication results of BFRP in N-SWSSC solution with Model 2 and Model 3 are exactly same.

The original tensile strength data of GFRP bars are extracted from the published literature [14]. The detailed prediction procedures of GFRP bars at different temperatures in different locations in Canada with two models just follow those of BFRP bars shown above.

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